Memorised Blades: data leakage inflates wind-turbine blade defect benchmarks, and tuning on them rewards it

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Abstract

Public benchmarks for wind-turbine blade defect detection report 85–92 mAP50. We audit the three most used, WTBD, Wind Surface Defect and WTBlade-Defect, with image retrieval, geometric verification and pixel correlation. We find that 78%, 84% and 92% of their official test images have a near-copy or same-scene image in the training split, because the datasets were augmented or tiled before a random split. A generic foundation detector (RF-DETR-M) with a standard recipe scores 90.4 mAP50 on WTBD's official split and 87.2 on WTBlade-Defect's, above the published BladeYOLO on the latter (85.8), though below it on Wind Surface (81.0 vs 86.5). The same runs lose 24 and 40 points on the Wind Surface and WTBlade-Defect test images without a training twin: on WTBlade-Defect they score 98.2 on copies of training frames and 50.4 on images with no twin. Trained with the same recipe on a scene-grouped split of the same WTBD images, the detector falls from 90.4 / 61.2 to 55.3 / 33.2 mAP50 / mAP50-95. A single step of recipe tuning against the official numbers raised the score on leaked images and lowered it on unseen ones on all three datasets, so leaderboard tuning on these splits rewards memorisation. We release two leak-free benchmarks with scene-level intervals. On them, frozen DINOv3 features account for the whole advantage of a modern YOLO variant at every label budget, and its deformable, selective-scan and loss components add nothing measurable. Fine-tuned RF-DETR leads on WTBD-group but not on WTBlade-group. We measure an annotation self-consistency ceiling of 84–87% on exact copies, a factor-of-two dependence of low-label results on which scenes are labelled, and a failure of conformal recall guarantees calibrated on seen scenes (31.5% missed against a promised 10%). Finally, more data from other sources does not replace target labels: a detector trained on 26,065 twin-free images from other datasets reaches 17.7 class-agnostic AP50 on WTBD-group test, against 72.9 for 745 target images, and 2.4 against 51.7 on WTBlade-group. A ledger of our pre-registered method tests, most of them failures, shows that the usual add-ons do not close the leak-free gap.

1 Introduction

Blade inspection by drone is one of the more mature applications of defect detection, and the published numbers suggest a nearly solved problem. BladeYOLO [1] reports 86.5 / 57.4 mAP50 / mAP50-95 on Wind Surface Defect and 85.8 / 74.6 on WTBlade-Defect, and other recent architectures report similar levels [2–4]. This work began as an attempt to reproduce one of those results and build on it. The closest faithful reimplementation we could build stopped at about 52.6 / 28, and our next model reached 78.2 / 46.9, until we found that three-quarters of the validation images had a near-identical twin in the training set. That finding changed the question from "how do we beat this number?" to three others:

1. What do these benchmarks measure?
2. How much of a reported score is memorisation of scenes seen in training?
3. Once leakage is removed, what actually helps, and what limits the score?

We answer them with an end-to-end audit and about two months of controlled experiments. Our contributions are measurements, not a new architecture. We tried to find a method that survives a second dataset, and did not; § 9 reports those attempts.

- **A leakage audit of the public blade benchmarks** (Table 1, Fig. 3). A literature check found no earlier audit of these datasets. The check was not exhaustive.
- **The size of the inflation**, on the same detector and recipe: leaked against no-twin test images (Table 3), and official against scene-grouped splits (Table 4, Fig. 10).
- **Evidence that tuning on leaky benchmarks selects for memorisation** (Table 5, Fig. 8). This analysis is post hoc and covers one tuning step.
- **Two leak-free benchmarks**, WTBD-group and WTBlade-group, with audited splits, scene groups and a scene bootstrap (§ 4), and baselines for RF-DETR, a DINOv3-fused YOLO and stock YOLO12-L (Table 7).
- **Controlled ablations** that attribute the low-label and full-label advantage of a DINOv3-fused YOLO to the frozen features alone (Table 8, Fig. 12).

- **Measured ceilings**: annotator self-consistency, box granularity, subset luck and error anatomy (§ 7).
- **A leave-one-dataset-out test** with twin-free pools (§ 8), and a recall-guarantee failure caused by leaked calibration (Table 6).
- **A ledger of the method tests we ran or formally gated**, including the failures (Table 21), and an account of how our own early numbers were wrong (§ 10).

The authors of the papers we discuss followed common practice, which is to augment and then split at random. Our argument is about the protocol, not about them. We cannot say that any specific published number is due to memorisation, because those splits were not published.

Fig. 1 shows how the parts of the study fit together.

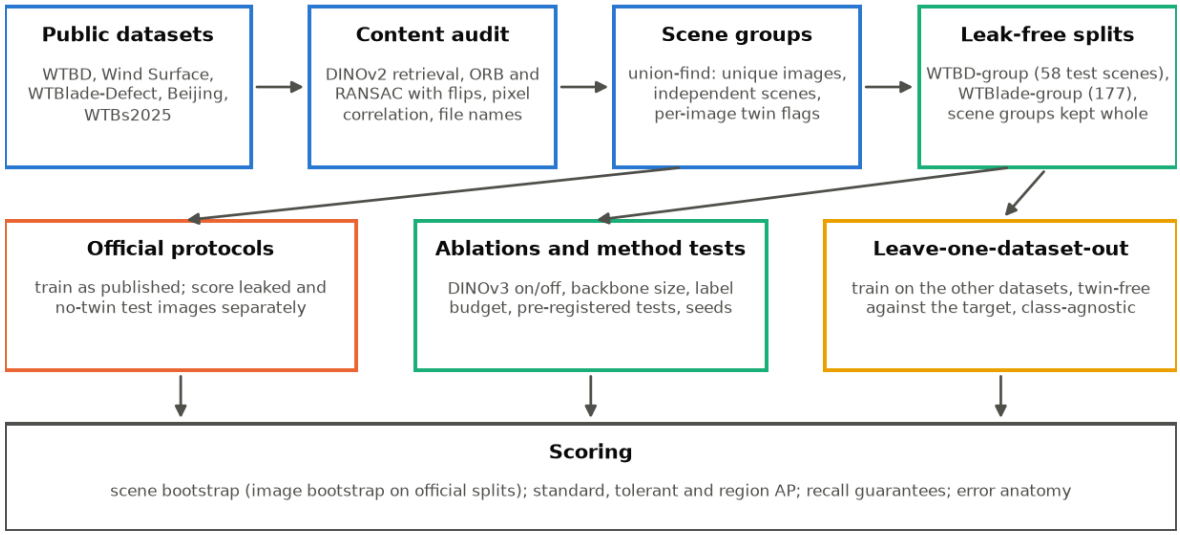


Figure 1. Overview of the study. The audit produces scene groups and per-image twin flags. These define the leak-free splits and the leaked and no-twin subsets of the official test sets. All experiments share the same scoring.

2 Related work

Blade defect detection. Most work adapts YOLO variants with attention, wavelet, deformable or Mamba-style modules [1–4], and several build on the DTU NordTank drone inspection imagery [5, 6]. BladeYOLO fuses frozen DINOv3 ViT-S/16 features into YOLOv12-L [1, 7], close in spirit to DINO-YOLO in civil engineering [8]. Data-descriptor papers have added new sets, notably WTBD [9] and WTBs2025 [10]. Training-free language-model inspection [11] and box-to-mask supervision [12] are recent directions, and we have not tested those specific claims on a leak-free split.

Leakage and near-duplicates. Near-duplicates between train and test are documented for CIFAR [13], geospatial mapping data [14] and more generally as a source of irreproducible machine-learning results [15]; and a fresh test set can show lower accuracy than a reused one [16]. Long-tail memorisation is a mechanism by which rare test examples are classified correctly through similar training examples [17]. We apply this line of work to blade defect benchmarks, and add a same-detector measurement of what the leak buys and what tuning on it rewards.

Foundation detectors. RF-DETR [18] fine-tunes DINOv2 [19] end to end and is strong on Roboflow100-VL [20]. DINOv3 [21] offers frozen dense features. Fine-tuning can distort pretrained features under distribution shift [22].

Label noise and granularity in evaluation. TedEval [23] handles split and merged instances in text detection. Box-annotation noise has been studied in detection evaluation [24] and in repeated labels [25], and box regression can model its own uncertainty [26]. We measure the same kind of ambiguity from accidental repeated labels: independently annotated copies of the same image.

Calibration and subset variance. Conformal risk control [27] gives distribution-free guarantees under exchangeability, and has been applied to detection [28]. Few-shot detection results vary strongly with the sampled shots [29]; we measure that variance in a realistic low-label setting and test label-free selection [30].

Multi-dataset and domain generalisation. Multi-dataset detection with partitioned label spaces [31] and training-domain model selection [32] inform our G27 and G28 designs. Copy-paste [33] is a known augmentation, and we test it only as a recipe component.

3 Datasets and the leakage audit

3.1 Datasets

We work with the official classes of each dataset, because the taxonomies conflict (below). Fig. 2 shows random labelled examples and Table 1 summarises the audit.

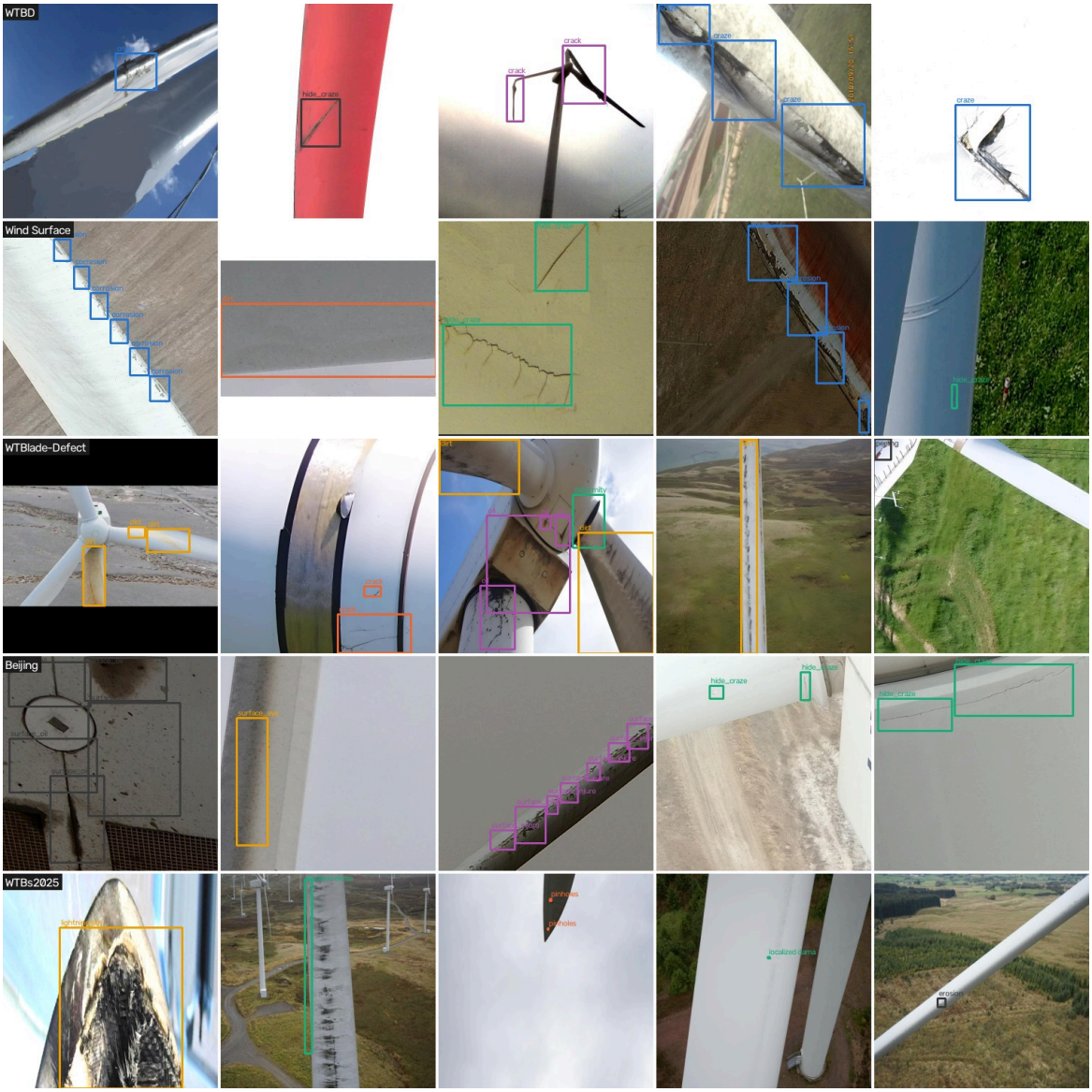


Figure 2. Random labelled training images from each dataset (fixed seed; the first three in each row are required to show a class not yet shown). Colour marks the class within a row. Note the differences in viewpoint and box style, from close-ups with one box (WTBD) to chains of small boxes along an edge (Wind Surface, Beijing) and very small boxes (WTBs2025). Black squares in WTBlade-Defect are the Roboflow export's offline cutout augmentation. Source licences are in § 13.

Table 1. The three benchmarks and their official splits. Unique images and scenes are from the content-based audit. A *twin* is a near-copy or same-scene image among the images the run trained on.

dataset	files	unique images	independent scenes	official split (train / val / test)	test images with a training twin	how the twin arises
WTBD [9]	1,065	597	~550	745 / 159 / 161	**125 / 161 (78%)**	duplicate and augmented copies of one photo, split at random (seed 42)
Wind Surface Defect [2], used by BladeYOLO	3,570	954	**81**	2,856 / 714 (val = test)	**600 / 714 (84%)** pixel twins (corr ≥ 0.99)	1,785 originals plus one augmented copy each (550 pairs split across train and val); up to 44 tiles per drone photo (386 of 882 photos on both sides)
WTBlade-Defect [34], used by BladeYOLO	10,660	1,146	803	9,010 / 1,097 / 553	**499 / 542 (92%)** same scene; **360 (66%)** copies of a training frame	about 3 Roboflow augmentation copies per source frame; consecutive drone-video frames

3.2 Audit method

Files cannot be compared by name or hash, because the copies are flipped, rotated, recoloured or re-encoded. We combine several signals:

1. **Retrieval:** DINOv2 embeddings [19], cosine nearest neighbours.
2. **Geometric verification:** flip-aware ORB matching with a RANSAC similarity transform, at least 25 inliers.
3. **Pixel twins:** correlation at 48 px over the eight flips and rotations, at least 0.99.
4. **Provenance from file names:** Roboflow source prefixes, `DJI_xxxx_r_c` tile coordinates, and `_0` / `_1` pairs.
5. **Grouping:** union-find over all of the above, with DINOv2 cosine ≥ 0.93 counted as same scene. Deduplication is non-chaining, so that a chain of weak matches cannot merge unrelated images.

Example sheets were checked by eye for every dataset and outcome. On Wind Surface the first pixel comparison, flips only, found 532 of 714 validation images (75%); adding rotations and restricting to the images actually trained on gives 600 (84%). The Wind Surface “no twin” subset has no near-identical copy in training, but most of it comes from drone photos with other tiles in training, so it is cleaner rather than leak-free.

Fig. 3 shows test images and their training twins, drawn at random.

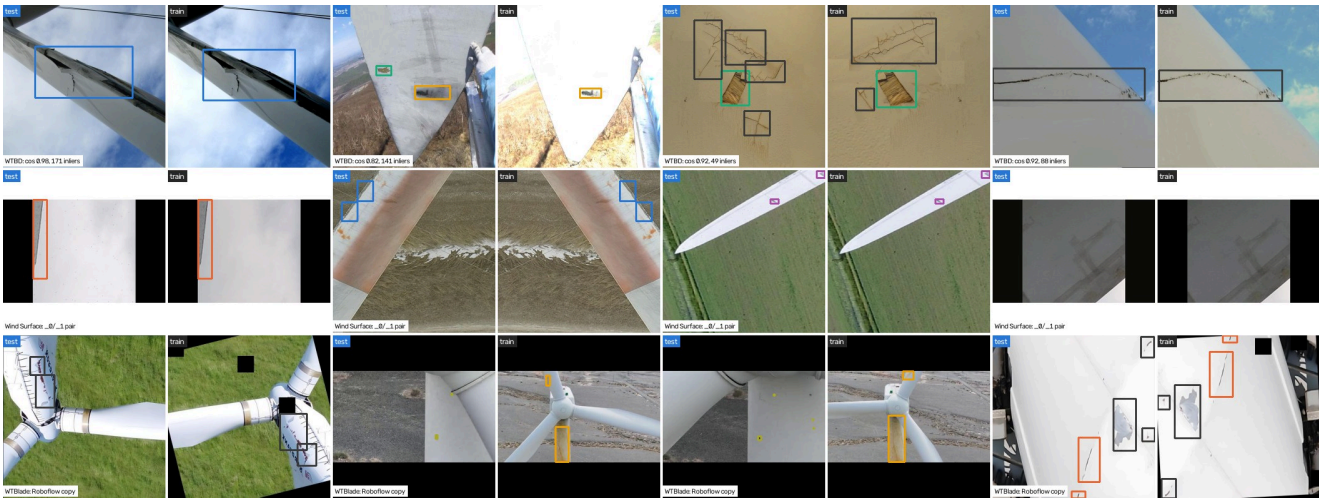


Figure 3. Official test images (blue tag) next to their training twins (black tag), drawn at random with a fixed seed from the leaked subsets. Labels are drawn as supplied. Row 1: WTBD, with DINOv2 cosine and ORB inliers. Row 2: Wind Surface `_0` / `_1` pairs split across the partition. Row 3: WTBlade-Defect Roboflow copies of one source frame.

3.3 What the file counts hide

File count is a poor guide to the information in a dataset (Fig. 4). Wind Surface’s 3,570 files come from about 81 independent scenes, and all `surface_eye` boxes (412 in the raw files, 103 after deduplication) come from tiles of a single photo. Any `surface_eye` score on the original split, including published ones, was measured on tiles of a training photo. WTBD has far fewer files and far more independent scenes per class. Three more findings limit what can be merged:

- **Same-named classes do not match across datasets.** Mutual crop retrieval between same-named classes is at most 0.33, apart from the two Beijing exports, which are the same data.
- **Datasets overlap under other names.** WTBs2025 shares 1,499 images with WTBlade-Defect under different class names (peeling → erosion). Beijing contains WTBD test photos.
- **Licences are not what the export pages say.** "NORDTANK" is legible on many WTBlade-Defect nacelles, so the set contains DTU imagery [5], which is CC BY-NC, while its Roboflow page states CC BY 4.0. It also contains synthetic copy-paste images.

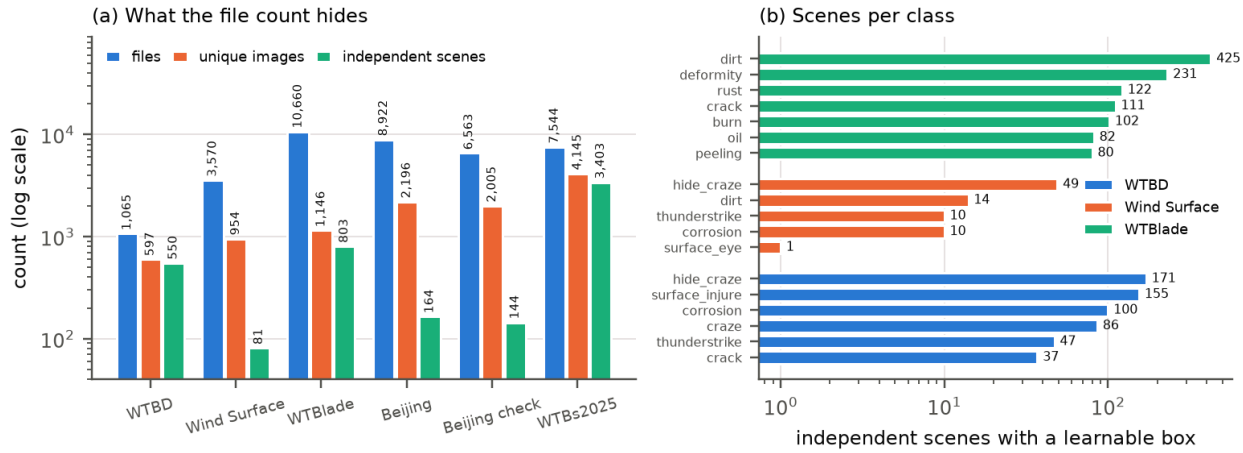


Figure 4. Effective size. (a) Files, unique images and independent scenes per dataset. (b) Independent scenes with a learnable box, per class. A single scene supports Wind Surface's `surface_eye`.

The same augment-then-split pattern appears in other published work: 130 DTU images expanded to 780 and then split 624 / 78 / 78 [4], and CLAHE copies with a random 60 / 40 split of repeated shots of one turbine [3]. BladeYOLO's WTBlade numbers use its authors' own 9:1 re-split of 5,326 images, which is not public [1].

4 Leak-free benchmarks, models and evaluation

4.1 Splits

WTBD-group. The 1,065 WTBD images are split by scene into 745 / 159 / 161 images (the official sizes), with 1,109 / 240 / 235 boxes and 224 / 60 / 58 scene groups. The split was audited until no held-out image matched train under DINOv2 and ORB verification. Fractions of 25% and 50% are nested subsets of training scenes.

WTBlade-group. From 10,527 WTBlade-Defect images (133 that match WTBD are left out, so the benchmarks share nothing), we form 841 groups from 2,791 source frames, verified pairs and DINOv2 neighbours. A class-stratified 70 / 15 / 15 split over groups converged with no held-out or train match after 11 rounds of the audit. We keep one image per source frame, giving 2,136 / 323 / 332 frames, 5,237 / 729 / 748 boxes, and 177 test scenes, with all seven classes in every split.

A residual leak in our own splits. While building a later pool we added the pixel test to our audit and ran it on these splits. It found copies that DINOv2 and ORB had missed: 6 of 161 test and 6 of 159 validation images on WTBD-group, and 1 of 332 and 1 of 323 on WTBlade-group (Fig. 5). All six WTBD-group test pairs are real copies by eye, re-exposed, darkened, tinted or rotated. Rescoring 116 cached WTBD-group runs without those images changes mAP50-95 by −0.62 on average (range −1.09 to +0.33) and mAP50 by −0.70, and 20 WTBlade-group runs by ±0.03. Our leak-free numbers are therefore inflated by under a point. We kept the splits unchanged, so that every comparison stays like for like, and report the G27 and G28 decisions both ways (§ 9, § 8). Neither changes. So a leak-free split needs the pixel test next to scene grouping, and "audited" is only as good as the audit.

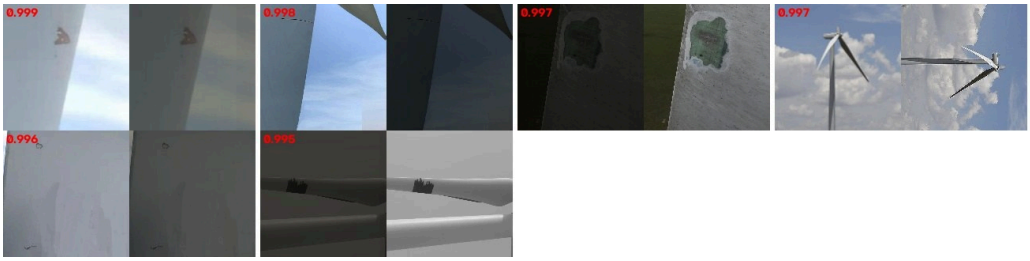


Figure 5. The residual leak in our WTBD-group test set: each pair is a test image and its pixel twin in train, with the correlation. Photometric copies defeat DINOv2 + ORB grouping.

4.2 Models and recipes

- **YOLO12-L:** COCO-pretrained, stock, CIoU loss, AdamW (lr 0.001, weight decay 0.01), cosine schedule, mosaic. 150 epochs on WTBD and 50 on WTBlade-group.
- **BladeYOLO-v2:** our model, on the same recipe. A COCO-pretrained YOLOv12-L [7] with three zero-initialised residual additions, so that training starts exactly at the pretrained detector: frozen DINOv3 [21] features fused at P3 / P4 / P5, deformable refinement at P3 / P4 [35], and four-direction selective scan at P5 [36, 37]; WIoU v3 loss [38]. 53.7 M parameters with ViT-S/16.
- **RF-DETR-M:** `rfdetr` 1.11.2 [18], fine-tuned end to end from its COCO checkpoint. Batch 4 × accumulation 4 (effective 16), cosine schedule, EMA checkpoint selected on validation. WTBD at 768 px for 100 epochs; WTBlade at 640 px for 60 epochs.

Runs used Kaggle Tesla T4 (fp16 mixed precision) and one A100 (bf16). All G27 seed-1 runs were on the A100 and seed-0 runs on Kaggle, so every pair of arms within a seed shares hardware.

4.3 Evaluation

Selection and reporting. Model selection uses validation only, and results are reported on test. Wind Surface has no separate test split, so we carve a seeded 10% of train for validation.

Evaluators. We use a 101-point COCO-style evaluator [39] (`tools/bootstrap_ci.py`). On identical predictions it agrees with Ultralytics' `ap_per_class` within 0.4 points on every run and subset, so gaps to published figures are not an artefact of the metric. Ultralytics' `val` reports differ by 1–2.5 points from cached predictions because of its rectangular-batch inference path. Tables state which is used.

Intervals. Test images of one scene are correlated, so on group splits we resample whole scenes, paired across models (2,000 replicates unless noted). Official splits have no scene groups, so those intervals are image-level and therefore narrower than they should be. All intervals are 95%.

Three box scores. *Standard* AP is one-to-one matching at IoU ≥ 0.5. *Tolerant* AP lets one prediction match several same-class labels whose union it fits, and ignores fragments of an already found defect. *Region* AP is pixel-level precision and recall of the union of predicted boxes against the union of labels (4 px grid), so it does not depend on how a defect is split into boxes.

Pre-registration. From late October, hypotheses were written down with their pass bars before any run or computation, and committed. After a bad result we allow one scope check, on whether any part is usable, and then discard the hypothesis without rescue runs. Post hoc analyses are labelled as such.

5 Official protocols: the scores and what they are made of

5.1 Our numbers against published

Table 2 gives our scores on the official protocols. A generic detector with a standard recipe, and no blade-specific modules, reaches the level of the specialised models on two protocols, but not on Wind Surface.

Table 2. Official protocols: this work against published. mAP50 / mAP50-95, single seed, selected on validation; intervals are image-level. BladeYOLO's WTBlade numbers use its own unpublished 9:1 re-split, so that row is not directly comparable.

protocol	model	mAP50 / mAP50-95 [95% CI]
WTBD-official (test 161)	YOLO12-L (ours, Ultralytics)	70.4 / 38.8
	BladeYOLO-v2 (ours, Ultralytics)	84.0 / 49.3
	RF-DETR-M 768, constant LR (G25)	90.5 [86.3, 94.4] / 60.2 [55.6, 65.3]
	RF-DETR-M 768, cosine (G26)	**90.4** [85.8, 94.7] / **61.2** [56.5, 66.1]
	RF-DETR-L 704, cosine (G26)	88.9 / 59.5
Wind Surface (val 714)	BladeYOLO [1]	**86.5 / 57.4**
	our best leaky BladeYOLO-L	78.2 / 46.9
	RF-DETR-M 640, 60 ep, constant (G25)	81.8 [78.0, 85.4] / 51.4 [48.3, 55.0]
	RF-DETR-M 640, 100 ep, cosine (G26)	81.0 [77.1, 84.9] / 53.3 [50.0, 57.0]
WTBlade-Defect (test 553)	BladeYOLO [1]	85.8 / **74.6**
	YOLOv12-L, as reported there	83.3 / 71.9
	RF-DETR-M 640, 25 ep, constant (G25)	87.4 [84.4, 90.9] / 52.5 [49.6, 56.2]
	RF-DETR-M 640, 40 ep, cosine (G26)	**87.2** [84.0, 90.6] / 55.1 [52.0, 59.1]

To our knowledge, the WTBD-official row is the best reported on that protocol; the dataset paper itself gives no detection baselines [9]. On Wind Surface, our copy has 3,570 images against the 3,790 reported by BladeYOLO, so the splits differ.

The mAP50-95 / mAP50 ratio. The ratio on WTBlade is 0.87 in the BladeYOLO paper and 0.86 for its YOLOv12-L baseline. RF-DETR's ratio on WTBlade copies of training frames is 0.71, on all official test images 0.63, and on no-twin images 0.34. Across all our leak-free runs it spans 0.41–0.60. A ratio of 0.87 sits above every one of these. This is consistent with near-duplicate copies falling on both sides of a random split, but it is not proof, because that split is not available.

5.2 Leaked and no-twin test images

Using the per-image flags from the audit, we score the same official runs on the two subsets (Table 3, Fig. 6).

Table 3. The same official runs on leaked and no-twin test images (COCO-style evaluator, mAP50 / mAP50-95 [95% CI], post hoc). Subset sizes (images / boxes): WTBD 125 / 185 leaked and 36 / 53 no twin; Wind Surface 600 / 1,169 and 114 / 158; WTBlade 499 / 1,235 and 43 / 91.

protocol, run	all	leaked	no twin
WTBD-official, YOLO12-L	69.0 / 37.7	69.7 / 38.5	70.0 / 35.2
WTBD-official, BladeYOLO-v2	81.5 / 48.1	82.6 / 48.9	81.5 / 47.8
WTBD-official, RF-DETR-M 768 cosine	90.4 / 61.2	91.5 / 63.9	81.6 [69.1, 93.2] / 49.9 [40.4, 61.1]
Wind Surface, RF-DETR-M cosine	81.0 / 53.3	83.3 / 55.1	**59.2** [49.0, 68.1] / **37.6** [29.9, 45.4]
WTBlade-Defect, RF-DETR-M cosine	87.2 / 55.1	90.8 / 59.4	**50.4** [38.5, 64.0] / **17.2** [13.5, 23.4]
WTBlade-Defect, copies of a training frame (360)		**98.2** [97.7, 98.8] / **69.8** [67.8, 72.5]	
WTBlade-Defect, same scene, not a copy (139)		65.7 / 30.1	

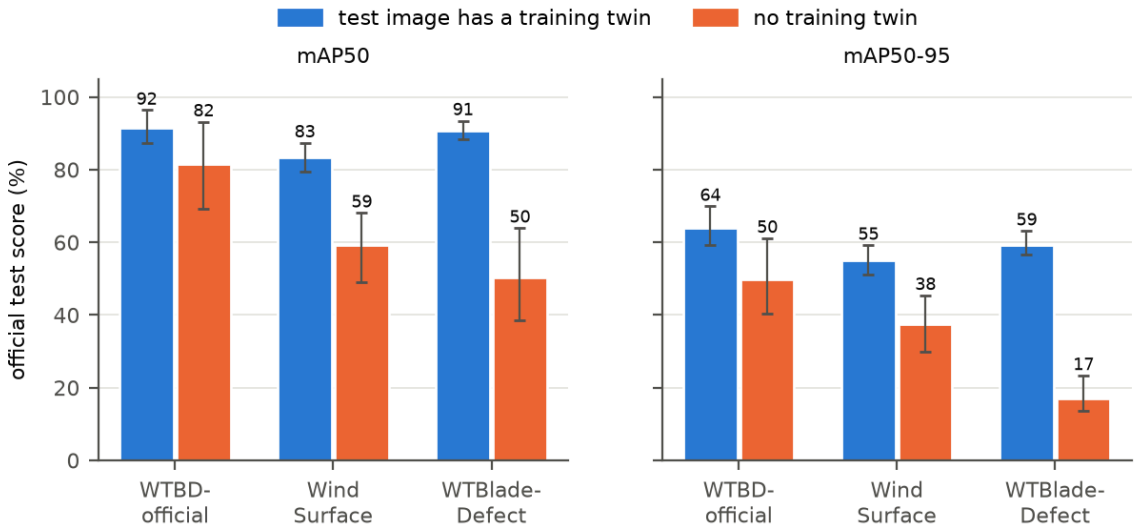


Figure 6. Official-protocol scores of the best RF-DETR-M run on leaked and no-twin test images. Bars show the COCO-style evaluator with image-bootstrap 95% intervals.

WTBlade-Defect's official number is mostly its copies: 98.2 on copies of training frames and 50.4 with no same-scene training image. The independently built WTBlade-group benchmark agrees from a different direction: its RF-DETR-M runs score 16.1–17.3 mAP50-95 (Table 7) against 17.2 here. Their 38.5–41.8 mAP50 is inside this small subset's interval of [38.5, 64.0].

WTBD's no-twin subset is too small to read on its own: it holds a single box each of craze, thunderstrike and crack. Corrosion is the only class with a clear drop (AP50 97 to 54). The split comparison below is therefore the stronger evidence for WTBD.

5.3 Same detector, only the split differs

Table 4 and Fig. 7 compare models trained on the same 1,065 WTBD images, official against scene-grouped.

Table 4. Same recipe, only the WTBD split differs. Test AP50 by class and overall mAP50 / mAP50-95. BladeYOLO-v2 and YOLO12-L use Ultralytics `val` and trained for 120 epochs on the official split and 150 on the group split. RF-DETR-M uses the COCO-style evaluator (seed 0), 768 px and 100 epochs on both splits, with an effective batch of 16 on both (2 × 8 and 4 × 4).

class	BladeYOLO-v2 official	BladeYOLO-v2 group	YOLO12-L official	YOLO12-L group	RF-DETR-M official	RF-DETR-M group
crack	99.1	38.1	85.4	38.4	95.3	61
craze	90.9	7.6	62.2	6.5	95.3	9
hide_craze	80.0	56.9	63.9	52.6	80.3	71
surface_injure	81.6	74.7	71.1	62.9	91.0	83
thunderstrike	99.5	98.0	93.8	41.8	94.3	64
corrosion	52.7	53.8	46.0	60.8	86.0	46
all	**84.0 / 49.3**	**54.8 / 24.5**	**70.4 / 38.8**	**43.8 / 20.6**	**90.4 / 61.2**	**55.3 / 33.2**

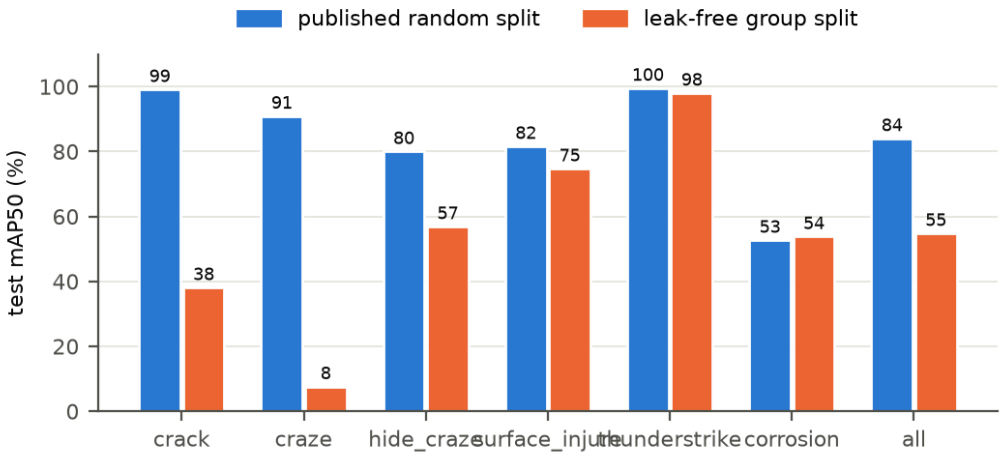


Figure 7. Inflation per class on WTBD for BladeYOLO-v2 and YOLO12-L, from the same recipe on the official and the group split.

The inflation is model-independent: +29.2 mAP50 for BladeYOLO-v2, +26.6 for YOLO12-L and +35.1 for RF-DETR-M, with mAP50-95 multiplied by 2.0, 1.9 and 1.8. Fig. 10 (c) adds WTBlade: 87.2 / 55.1 officially and 38.5 / 16.2 on WTBlade-group. That second pair is not a pure split comparison, because WTBlade-group also keeps one frame per source and 60 rather than 40 epochs.

5.4 Tuning on the official splits rewards memorisation

Between G25 and G26 we made one tuning step in the usual spirit, moving to a cosine schedule (and on two protocols, longer training) because the official numbers improved. Table 5 decomposes what that step bought (post hoc).

Table 5. What one tuning step bought. Paired image bootstrap, Δ mAP50-95 in points = G26 recipe − G25 recipe, same images. Interaction = Δ leaked − Δ no twin, with the subsets resampled independently.

protocol (change)	Δ all	Δ leaked	Δ no twin	interaction, P(>0)
WTBD-official (constant → cosine)	+1.0 [−1.2, +3.0]	+1.8 [−0.1, +4.0]	−8.1 [−13.4, +0.2]	+9.9 [+1.1, +15.8], 0.993
Wind Surface (60 ep constant → 100 ep cosine)	+1.8 [+0.5, +3.1]	+2.3 [+0.9, +3.8]	−1.5 [−4.2, +1.4]	+3.8 [+0.7, +7.0], 0.988
WTBlade-Defect (25 ep constant → 40 ep cosine)	+2.6 [+1.1, +3.9]	+3.5 [+2.0, +5.0]; copies +5.4 [+3.6, +7.0]	−4.0 [−7.2, +0.1]	+7.4 [+3.0, +11.1], 1.000

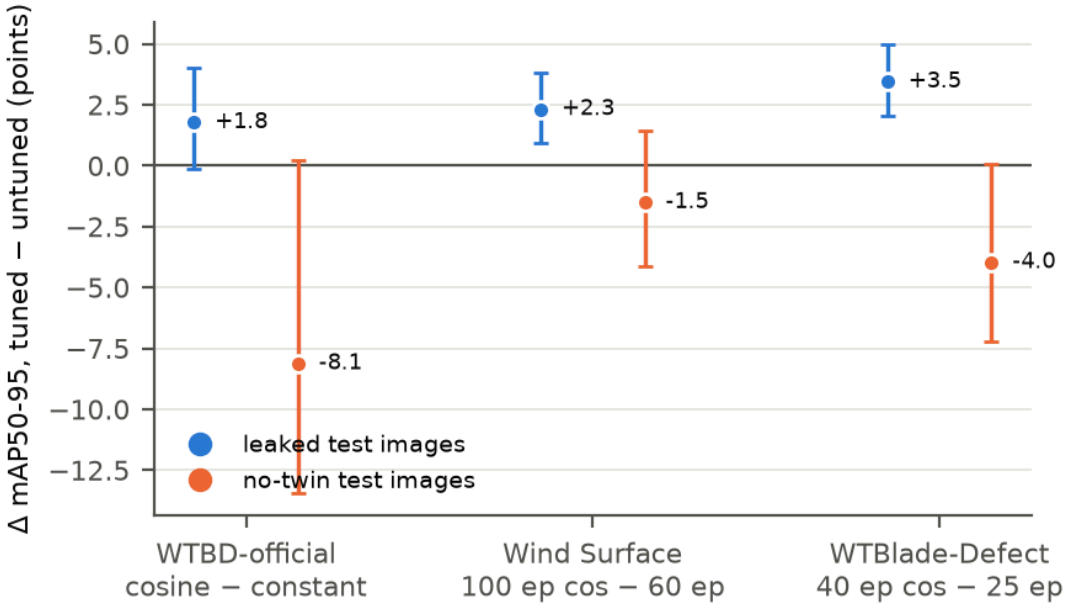


Figure 8. The tuning step of Table 5: gains sit on leaked images, and the same change lowers the no-twin score.

On every protocol the gain sits on leaked images, and the same change lowers the score on images without a twin. At mAP50 the change is −0.2 to −0.8 overall and −5.0 to −6.0 on no-twin images. The interaction excludes zero on all three. This applies to our own G26 numbers, which stay in Table 2 labelled as what they are. Two confounds remain: on Wind Surface and WTBlade-Defect the epoch count also changed (on WTBD only the schedule changed, with the same pattern), and this is one tuning step, not a study of tuning in general.

A second reading comes from the leaderboard itself. RF-DETR-M leads BladeYOLO-v2 on WTBD-official by +13.2 [+8.2, +17.4] mAP50-95 on all images, but by only +2.1 [−5.3, +19.8] on the 36 no-twin images, so that official lead is not established on unseen images.

5.5 Consequence for deployment: leaked calibration breaks recall guarantees

An inspection system should promise a bound on missed defects. Conformal risk control [27] picks a confidence threshold so that the expected share of missed defects is at most α. A defect counts as flagged if at least 50% of its box is covered by predictions above the threshold. We calibrate on seen or unseen scenes and always deploy on unseen scenes (Table 6, Fig. 9).

Table 6. Conformal recall guarantee, target ≤ 10% missed (BladeYOLO-v2, 500 draws of 100 calibration images, or scene groups as units).

calibrated on	deployed on	realised miss rate	draws violating
seen scenes (group train), image units	unseen scenes	**31.5%**	100%
seen scenes, scene units	unseen scenes	28.8%	100%
unseen scenes (group val), image units	unseen scenes	6.3%	0%
unseen scenes, scene units	unseen scenes	4.7%	1%
official val (leaky)	official test, images with a twin	12.6%	91%
official val (leaky)	official test, no-twin images	**25.9%**	100%

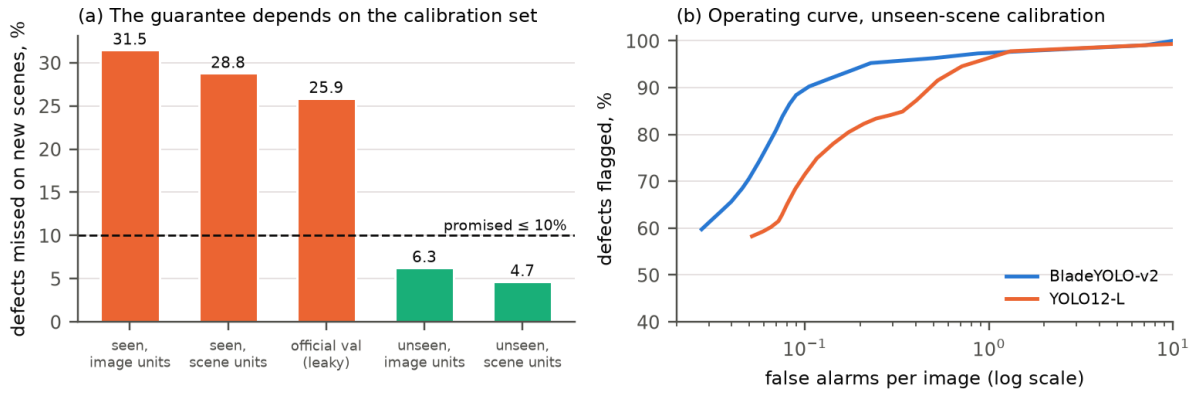


Figure 9. (a) Realised miss rate on new scenes, by calibration set. (b) Operating curve with scene units, calibrated on unseen scenes. At 0.1 false alarms per image BladeYOLO-v2 flags about 90% of defects on new scenes, YOLO12-L about 71%. One test set of 58 scenes; the curve has no interval.

Models are overconfident on what they memorised, so a threshold calibrated on seen scenes is too high for new data. The practical rule is that calibration data must be split by scene, as test data must. The honest calibration is cheap: 0.2 instead of 0.1 false alarms per image. With only 60 calibration and 58 deployment scenes from one site, the val and test groups are not perfectly exchangeable: YOLO12-L still misses 12.3% with scene-unit calibration on unseen scenes, violated in 71% of draws.

6 Leak-free results: what helps

6.1 The benchmark at full labels

Table 7, Fig. 10 and Fig. 11 give the benchmark, all with one evaluator.

Table 7. Leak-free benchmark, full labels, test mAP50 / mAP50-95, COCO-style evaluator on cached predictions. YOLO12-L on WTBlade-group trained for 50 epochs, reached its best validation at epoch 43, and rose almost linearly until then (Fig. 15 f), so its row may understate it. No paired interval was computed for the cross-architecture comparisons in this table, so treat them as point estimates.

model	WTBD-group (161 images, 58 scenes)	WTBlade-group (332 frames, 177 scenes)
YOLO12-L	40.0 / 18.5	32.2 / 13.3
BladeYOLO-v2 without DINOv3	31.8 / 15.2	not run
BladeYOLO-v2	56.1 / 24.4 (ViT-S)	41.8 / 18.0 (ViT-B)
RF-DETR-M, A0, seeds 0 / 1	55.3 / 33.2, 54.8 / 32.2	38.5 / 16.2, 41.0 / 16.9
RF-DETR-M + copy-paste, A1, seeds 0 / 1	51.5 / 30.7, 58.5 / 32.7	41.8 / 17.3, 39.3 / 16.5
RF-DETR-M + copy-paste + in-domain SSL, A2	49.6 / 27.4	40.1 / 16.1
RF-DETR-M + copy-paste + multi-dataset stage 1, A3	55.3 / 33.2	41.7 / 16.6

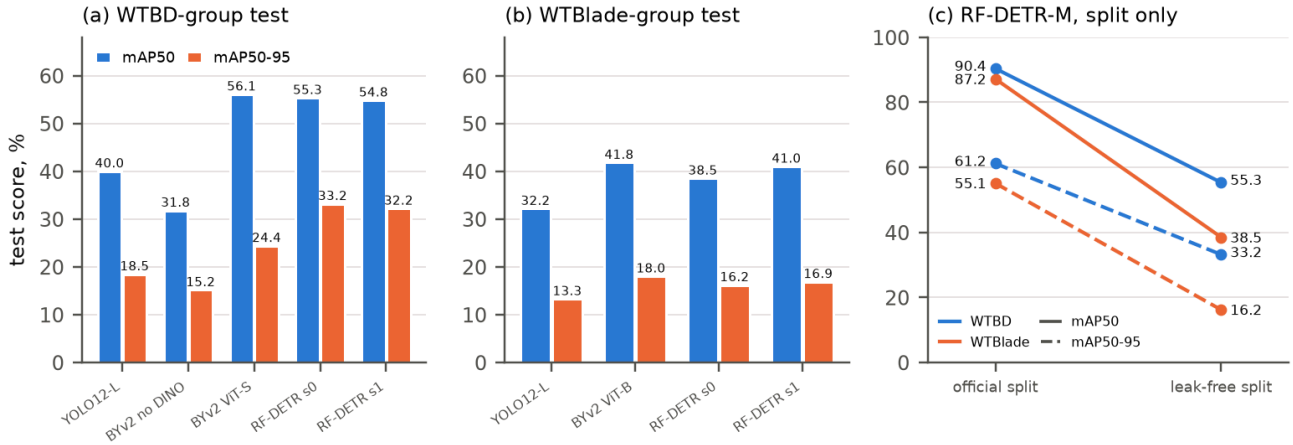


Figure 10. (a, b) Leak-free test scores by model (BYv2: BladeYOLO-v2; s0, s1: seeds). (c) RF-DETR-M on official and leak-free splits.

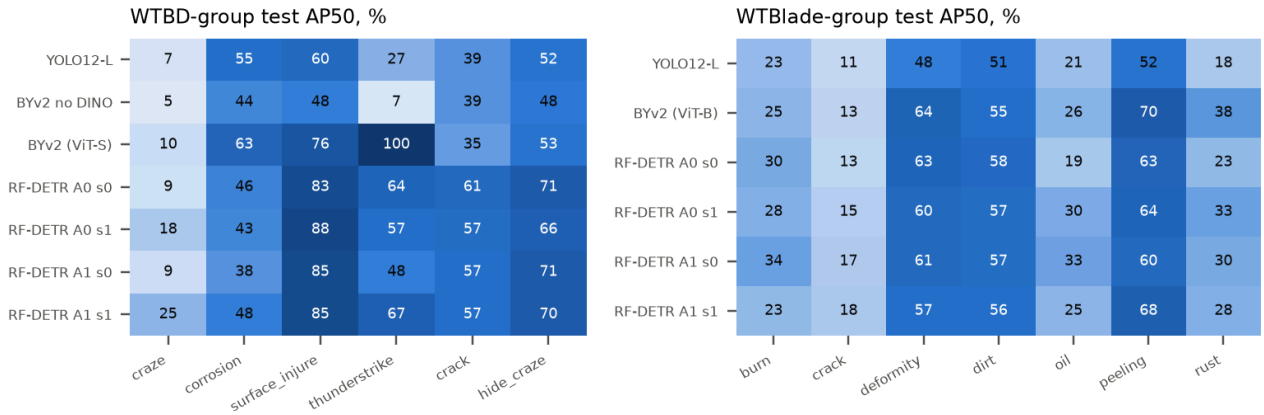


Figure 11. Per-class test AP50 on the leak-free benchmarks. Note the seed-to-seed swings within RF-DETR on rare classes (craze 9–25, thunderstrike 48–67), and BladeYOLO-v2's thunderstrike 100, which rests on 3 test scenes.

Four points stand out:

- **The ranking depends on the dataset.** RF-DETR-M leads on WTBD-group by 8–9 mAP50-95 points over BladeYOLO-v2, but on WTBlade-group the frozen ViT-B BladeYOLO-v2 (18.0) is ahead of every RF-DETR run (16.1–17.3). A "generic detector is strongest" claim holds on one benchmark.
- **On WTBD the RF-DETR gain is box quality, not finding.** mAP50 is level with BladeYOLO-v2 (55.3 and 54.8 against 56.1) while mAP50-95 is 8–9 points higher. This matches the extent analysis in § 7.
- **Classification costs points on RF-DETR.** Scoring A0 with classes merged gives 66.1 / 40.5 on WTBD-group and 50.1 / 21.3 on WTBlade-group (Table 15), about 11 AP50 points above its class-aware score. BladeYOLO-v2's class-agnostic and class-aware AP50 are about equal (56.3 against 56.1), while YOLO12-L shows a gap like RF-DETR's (48.5 against 40.0).
- **The earlier BladeYOLO-v2 win over YOLO12-L rests largely on thunderstrike.** The scene bootstrap gives +16.1 [+2.5, +28.4] mAP50 and +5.9 [+0.5, +16.9] mAP50-95, but without the two classes seen in at most 3 test scenes the

mAP50 gain is +5.2 [−2.7, +13.5].

6.2 Where the advantage comes from: DINOv3 ablation

We removed only the frozen DINOv3 features from BladeYOLO-v2, keeping deformable refinement, selective scan and WIoU, at 25%, 50% and 100% of the labelled scenes (Table 8, Fig. 12).

Table 8. DINOv3 ablation and learning curve, WTBD-group, test mAP50-95, scene bootstrap. Last row: stock YOLO12-L compared with our modules without DINOv3.

	25%	**50%**	**100%**
BladeYOLO-v2 (frozen DINOv3 ViT-S)	7.6 (3 seeds)	17.5	24.2 (2 seeds)
without DINOv3	3.7 (2 seeds)	8.8	15.2
YOLO12-L (stock)	4.1	9.9	18.5
DINOv3 gain	**+3.9 [+2.5, +10.1]**	**+8.8 [+4.0, +22.1]**	**+9.0 [+4.0, +20.5]**
same, 4 well-represented classes	+4.4 [+2.3, +11.1]	+6.1 [+2.2, +17.9]	+4.9 [−1.8, +10.3]
YOLO12-L − without DINOv3	+0.4	+1.1	+3.3 [−0.2, +13.1]

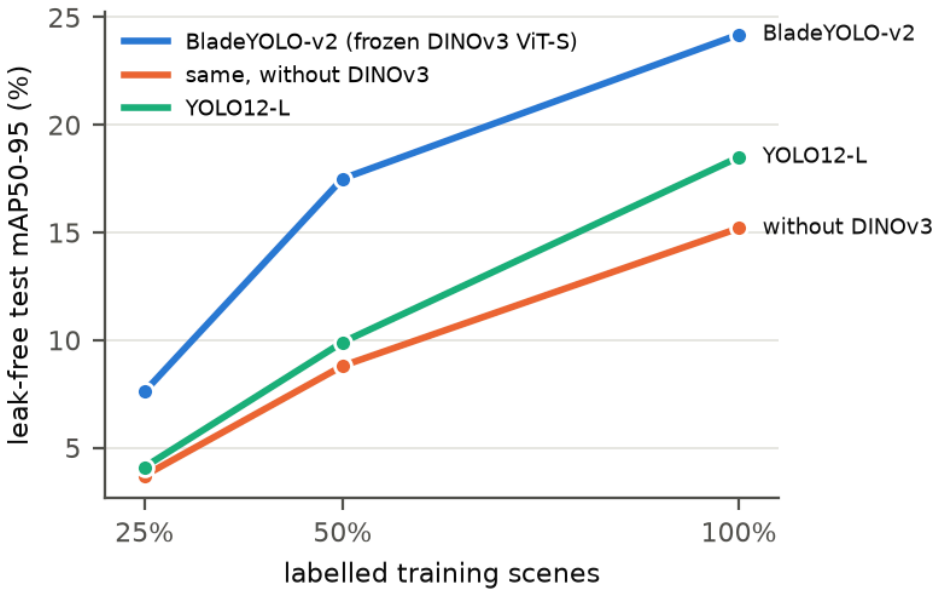


Figure 12. Learning curves with and without frozen DINOv3 on WTBD-group. The modules around DINOv3 do not put the model above stock YOLO12-L.

Without DINOv3, BladeYOLO-v2 equals or trails stock YOLO12-L at every data size. The advantage is therefore the features, not the architecture around them. We first read it as a low-data advantage; the 50% and 100% runs corrected that, since the absolute gain does not shrink with data. Only the relative gain does (2.0× to 1.6×). This claim concerns our components. BladeYOLO's exact Style Injection, WTConv and Cross-Mamba modules were reimplemented only before our code audit (§ 10), so we make no claim about them.

Backbone size at 25% (Table 9). A larger frozen backbone helps up to ViT-B, and ViT-L adds nothing further.

Table 9. Frozen DINOv3 size at 25% of WTBD-group scenes. Test scores by Ultralytics `val`; differences by scene bootstrap, mAP50-95 in points.

backbone	**test mAP50 / mAP50-95 per seed**	**difference**	**Δ mAP50-95 [95% CI]**
ViT-S/16, 3 seeds	23.2 / 8.2, 22.4 / 8.7, 18.4 / 6.8		
ViT-B/16, 2 seeds	25.2 / 10.2, 28.3 / 11.6	ViT-B − ViT-S	+2.8 [+0.4, +9.6]
ViT-L/16, 2 seeds	33.7 / 11.5, 30.1 / 10.9	ViT-L − ViT-B	+0.3 [−4.0, +3.1]
without DINOv3, 2 seeds	11.4 / 4.4, 8.1 / 3.1	without − ViT-S	−3.9 [−10.1, −2.5]

Fine-tuned against frozen foundation features, 25% labels, same subsets. RF-DETR-M beats BladeYOLO-v2 ViT-B by +7.3 [+2.3, +11.9] mAP50-95 (seed 0) and +8.3 [+4.5, +15.0] (seed 1), positive on 4 of 4 subsets at each seed. As a controlled comparison, freezing RF-DETR's encoder lowers test mAP50-95 from 18.9 to 12.3 on WTBD and from 13.2 to 8.7 on WTBlade. Input resolution (G24) adds +1.0 from 576 to 768 on WTBD (P 0.958) and −0.1 on WTBlade, whose frames are natively 640 px.

6.3 Training behaviour

Fig. 15 shows validation curves from every recent run. RF-DETR peaks early and then drifts down, which is why EMA checkpoint selection matters. In contrast, YOLO12-L on WTBlade-group rises almost linearly until about epoch 43 of 50. Fig. 13 shows training and validation batches, and Fig. 14 confusion matrices and a PR curve. In the matrices, columns are true classes. The bottom row shows missed labels (crack 18% on WTBD-group and 27% on WTBlade-group for YOLO12-L), the right column shows where false positives go, and the strongest confusions involve rare classes (YOLO12-L calls 54% of thunderstrike labels craze on WTBD-group test).

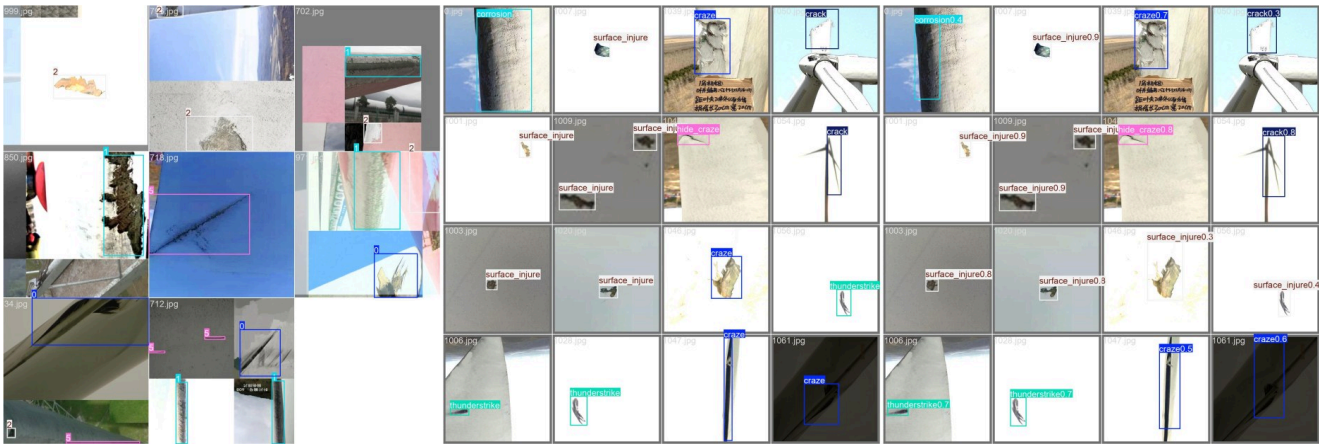


Figure 13. BladeYOLO-v2 on WTBD-group, from the training run. Left: a training batch after mosaic and flips. Middle: a validation batch with its labels. Right: the same batch with predictions.

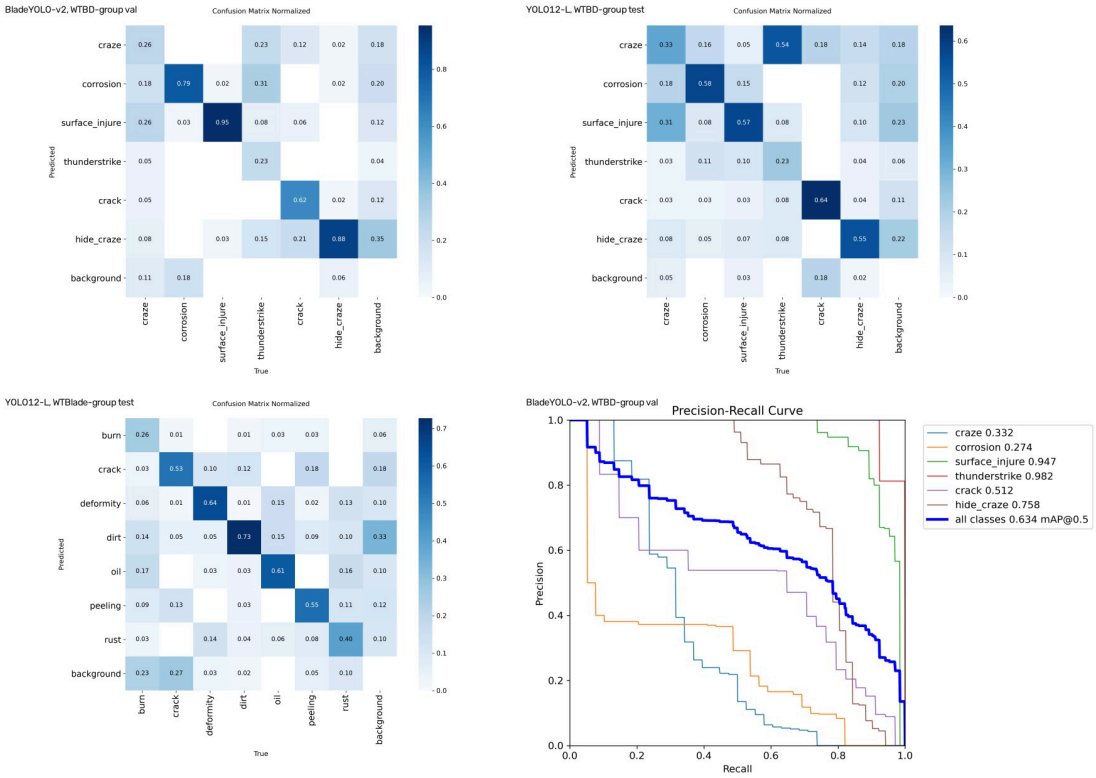


Figure 14. Diagnostics from the runs. Normalised confusion matrices: BladeYOLO-v2 on WTBD-group validation, YOLO12-L on WTBD-group test, and YOLO12-L on WTBlade-group test. Bottom right: PR curves by class of BladeYOLO-v2 on WTBD-group validation.

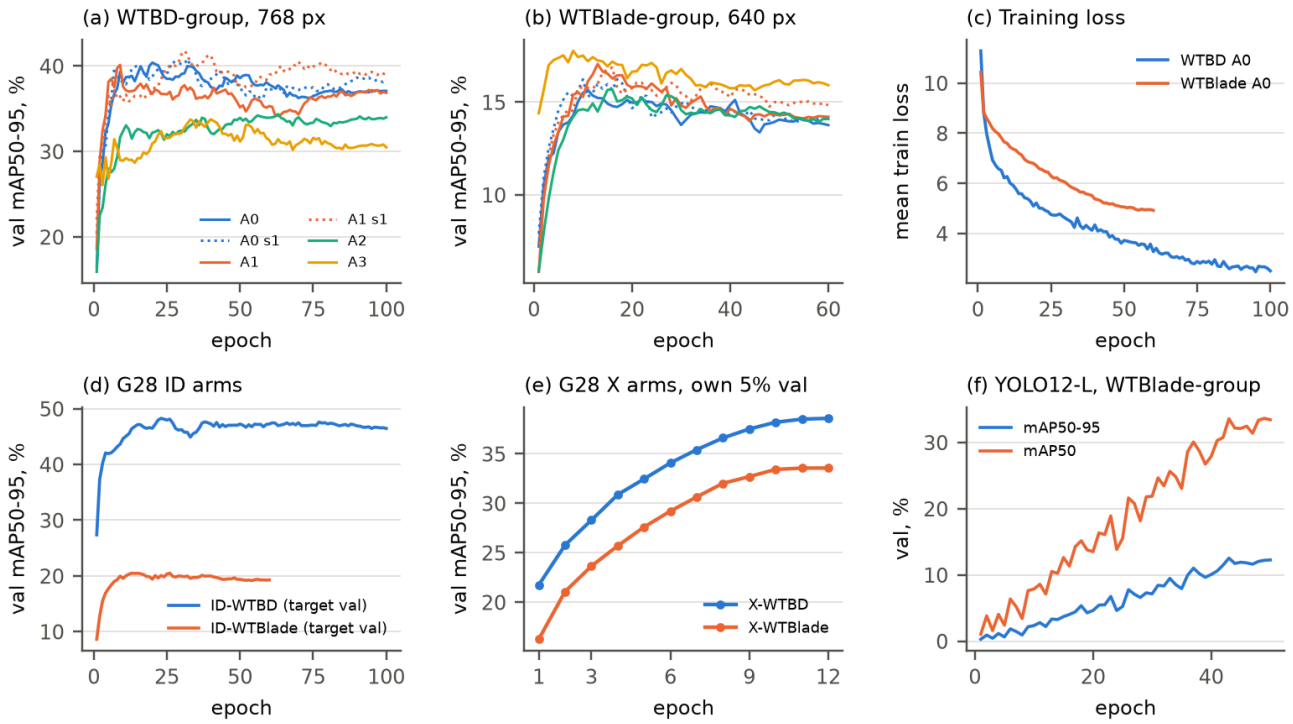


Figure 15. Training records. (a, b) Validation mAP50-95 of the G27 arms; dotted lines are seed 1. (c) Training loss of A0. (d) G28 ID arms on target validation. (e) G28 X arms on their own held-out 5%, not comparable with (d) since the images differ. (f) YOLO12-L on WTBlade-group.

7 Why leak-free scores are low

7.1 The annotators against themselves

WTBD's augmented copies were annotated independently, so aligned copies of an image give accidental repeated labels. For every verified pair of the same surface we project each label through the transform (flip-aware ORB + RANSAC similarity, ≥ 25 inliers, at least 8 inliers inside the box) and classify it as agree (same class, IoU ≥ 0.5), class conflict, split, loose or missing. Fig. 16 shows examples and Table 10 gives the rates.

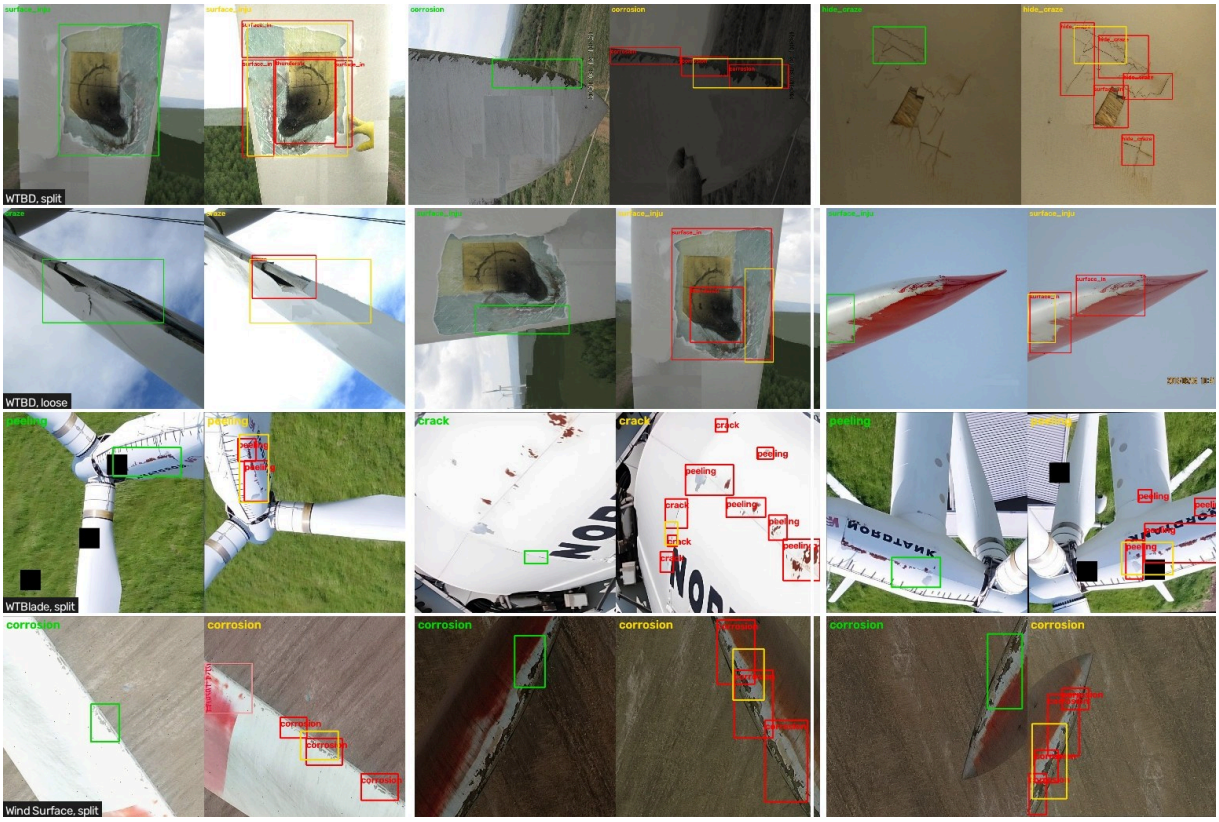


Figure 16. Annotation ambiguity on aligned copies. In each pair, left: a label (green). Right: its exact projection (yellow) and what the copy's annotator drew (red). Top two rows: WTBD. Third row: WTBlade-Defect. Bottom row: Wind Surface. The disagreement is about extent and how a defect is cut into boxes, not about the class. The pairs come from the audit's randomly ordered example sheets; "NORDTANK" is legible in the third row.

Table 10. Label self-consistency on axis-aligned copies of the same image (rotation within 3° of a multiple of 90°, flips included), where boxes map to boxes and every difference is an annotation difference. Rotated copies and other views of the surface agree less, but those strata also include geometric effects.

dataset	labels	agree	split	loose	missing	class conflicts	rotated copies	other views
WTBD	2,965	**87.3%**	4.6%	6.7%	1.5%	0	77.9%	64.1%
WTBlade-Defect	42,812	**86.0%**	2.5%	8.2%	3.1%	91 (0.2%)	80.7%	59.8%
Wind Surface	10,261	**83.8%**	9.2%	5.9%	1.1%	0	(18 labels)	44.2%

So 13–16% of labels disagree with an exactly aligned copy, in all three datasets, including a double-annotated one (κ = 0.897 reported by its authors [9]). Almost none of it is class. The amorphous classes are the least consistent (Fig. 17). Counts are per label occurrence and concentrate in a few heavily duplicated scenes. On WTBlade the "missing" rate is mostly a matching artefact of rotating rotors in video (1.1% without dirt). Most of its 91 class conflicts are pasted synthetic patches labelled peeling in one copy and crack in another.

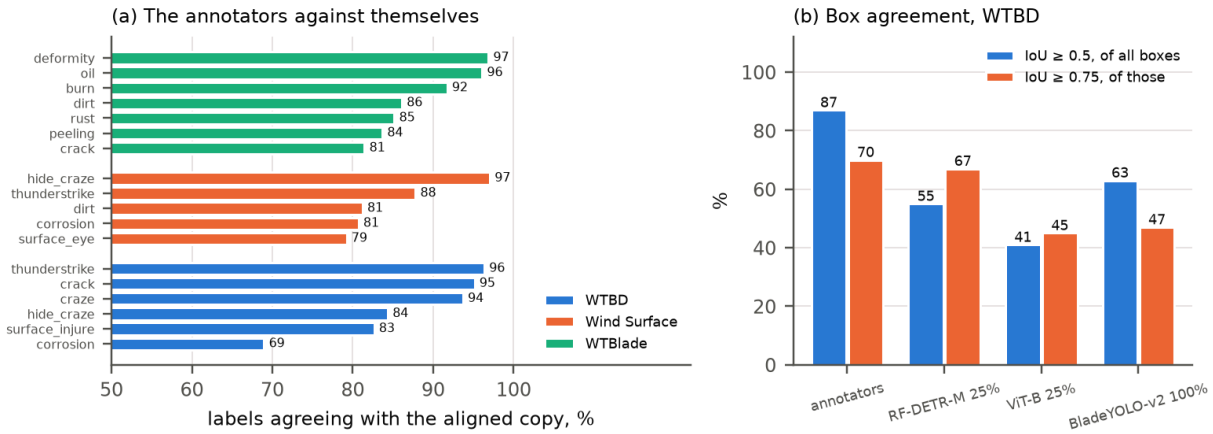


Figure 17. (a) Per-class agreement on aligned copies. (b) Box agreement on WTBD: how often an independent annotator's box, or a detector's best box, reaches IoU 0.5, and of those, how many reach 0.75. Conf ≥ 0.1 for detectors. RF-DETR-M at 25% labels matches the annotators on extent once it finds a defect (0.67 against 0.70; median IoU 0.83 for both).

7.2 Box granularity

On WTBD-group, tolerant AP50 exceeds standard AP50 by +11.9 for BladeYOLO-v2 (56.1 → 68.0) and +12.4 for YOLO12-L (40.0 → 52.4). Most striking is craze, from 9.9 to 66.1. All 15 craze test images are rotated, flipped or recoloured copies of one long leading-edge split, annotated as 2–3 segment boxes (Fig. 18). The same effect holds for RF-DETR scored with classes merged: AP50 rises from 66.1 to 79.7 on WTBD-group and from 50.1 to 61.2 on WTBlade-group (Table 15).

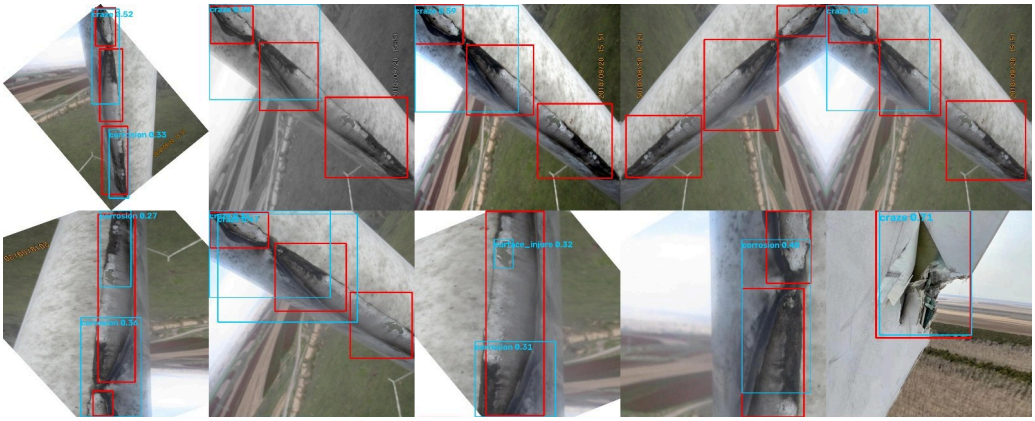


Figure 18. All 15 WTBD-group craze test images come from one scene. Red: labels, split into segments by the annotator. Blue: BladeYOLO-v2 predictions. The model often draws one box where the label has several, which counts as an error at IoU 0.5 and explains most of the "craze failure".

Table 11 gives the per-class view for BladeYOLO-v2.

Table 11. Three box scores on WTBD-group test, in percent (500 scene replicates). Per class: BladeYOLO-v2. The official-split rows use the same scores, and the official-minus-group gap shrinks as the metric becomes split-invariant (BladeYOLO-v2: +25.4, +20.9, +16.8).

	standard AP50	**tolerant AP50**	**region AP**
BladeYOLO-v2, WTBD-group	56.1 [48.3, 79.1]	68.0 [60.4, 83.1]	60.3 [55.2, 74.3]
YOLO12-L, WTBD-group	40.0 [33.9, 67.8]	52.4 [46.5, 74.0]	44.1 [40.2, 59.2]
difference	+16.1 [+2.3, +28.4]	+15.6 [+0.1, +23.7]	**+16.2 [+7.7, +23.6]**
BladeYOLO-v2, WTBD official	81.5	88.9	77.1
YOLO12-L, WTBD official	69.0	80.0	66.6
craze	9.9	**66.1**	42.7
corrosion	62.6	67.6	68.4
surface_injure	75.9	81.6	68.9
thunderstrike	100.0	100.0	68.2
crack	35.2	35.6	49.1
hide_craze	53.0	56.8	64.5

Two limits apply. We hypothesised that part of the leakage inflation is memorised box splitting, and tested it: the tolerant gain is not smaller on test images with a training twin, so the hypothesis is not supported. And training a detector to tolerate splitting does not help overall: blanket split-tolerant training is not an improvement (tolerant AP50 +1.5 [−11.1, +6.4]), and class-selective tolerance raises corrosion by +9.3 [−0.3, +24.6] while the mean moves by −0.4. So the ambiguity limits what can be measured more than what can be trained away.

Annotators as a ceiling. Corrosion copies agree on only 69% of labels and models score near that (BladeYOLO-v2 AP50 54–63), so more corrosion "improvement" is largely unmeasurable with these labels. Crack copies agree on 95% while the YOLO family scores 35–39, so crack is a genuine model weakness there.

7.3 Subset luck

At a fixed budget of 25% of scenes, which scenes are labelled matters more than any model effect we measured (Fig. 19). Across 8 random draws, with the same model, recipe and seed, ViT-B test mAP50-95 spans 9.0–19.6 (×2.2), and without DINOv3 it spans 2.6–7.2. The DINOv3 gain holds on 8 of 8 draws (+7.1 [+5.1, +12.6]), and draw quality carries over between models (Spearman ρ 0.71). Yet no label-free subset property we tested predicts it (H7, G16, H20). Label-free scene selection by patch coverage beats random by +5.2 [+1.9, +10.5] but does not separate from FreeSel [30] (+2.1 [−1.2, +9.1], P 0.917, validation opposite).

Table 12 gives the selection runs at the same 25% budget. Selecting by class balance does worst of the five arms, which suggests that which scenes are labelled matters more than matching class counts.

Table 12. Selection of which 25% of WTBD-group scenes to label: 18 ViT-B runs at the same budget, training seed 0 unless noted. Random draws per run: 11.2, 13.5, 11.8, 12.3, 9.0, 10.0, 12.4, 19.6.

arm (runs)	test mAP50	test mAP50-95	val mAP50-95	rank among 8 random (test / val)
patch coverage, label-free (4)	40.6	**17.7**	23.6	2 of 9 / 3 of 9
FreeSel-224 (2)	42.9	15.6	**27.1**	2 of 9 / 1 of 9
FreeSel-640 (2)	34.4	14.5	26.6	2 of 9 / 2 of 9
random (8)	31.3	12.5	21.6	
class-matched, uses labels (2)	25.3	10.4	19.7	7 of 9 / 7 of 9

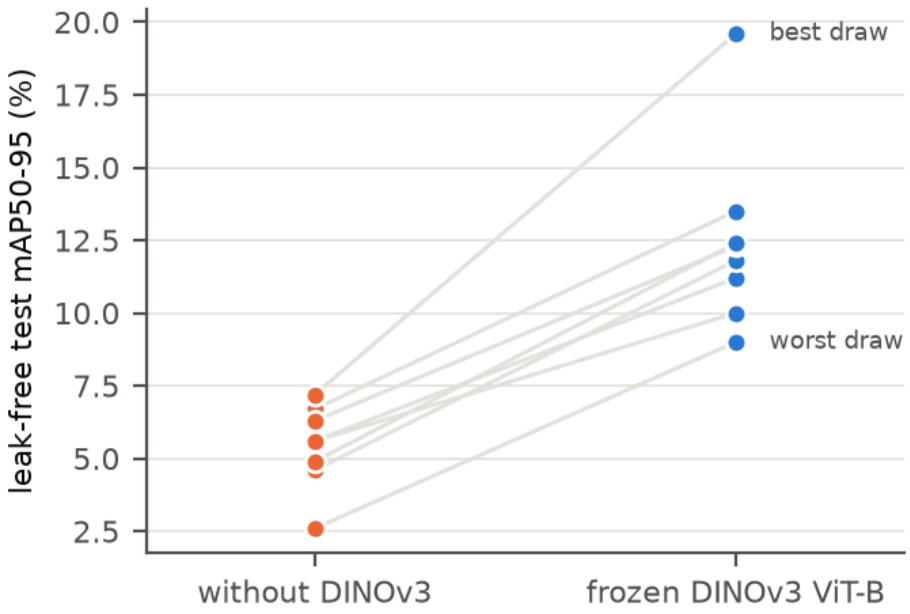


Figure 19. Subset luck: test mAP50-95 across random 25% scene draws, same model, recipe and seed.

7.4 Error anatomy

Scarce labels fail differently by detector family (Fig. 20, in the style of TIDE [40]). For each test box we ask whether a same-class box at IoU ≥ 0.5 is among the outputs and how it is scored.

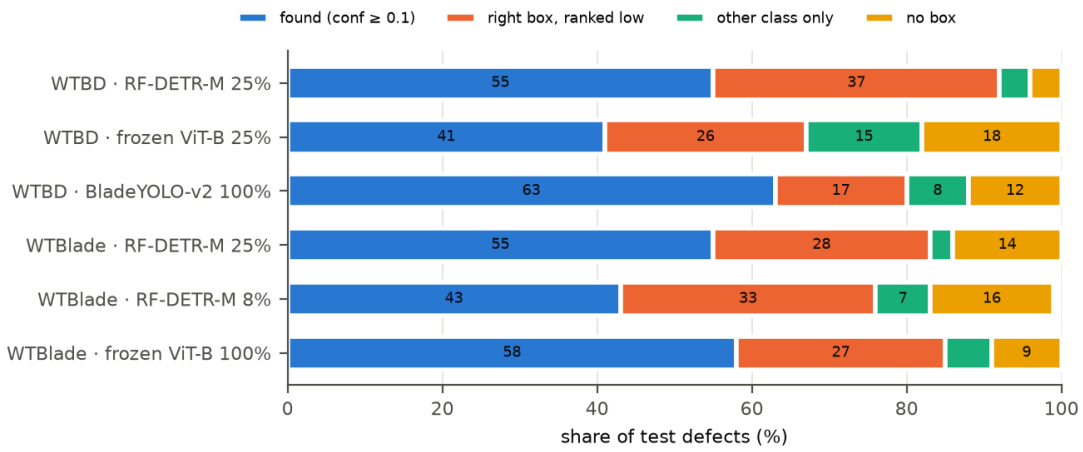


Figure 20. Error anatomy. Hit: same-class box at IoU ≥ 0.5 with confidence ≥ 0.1. Ranked low: such a box exists only below 0.1. Other class: only other-class boxes at IoU ≥ 0.5. No box: nothing at IoU ≥ 0.5.

Table 13 gives the numbers and Table 14 the extent comparison.

Table 13. Error anatomy, as a share of test boxes. Candidate recall = hit + ranked low. YOLO-family runs keep fewer boxes after NMS (about 50–80 per image) than RF-DETR (about 300), so recall within each image's top 50 boxes is also given.

dataset	detector	hit	ranked low	other class	no box	candidate recall	in top 50
WTBD	RF-DETR-M 25% (4 subsets)	0.55	**0.37**	0.04	0.04	**0.92**	0.81
WTBD	ViT-B 25% (4 subsets)	0.41	0.26	0.15	0.18	0.67	0.66
WTBD	BladeYOLO-v2 100% (2 seeds)	0.63	0.17	0.08	0.12	0.80	0.80
WTBlade	RF-DETR-M 25% (4 subsets)	0.55	0.28	0.03	0.14	**0.83**	0.72
WTBlade	RF-DETR-M 8% (4 subsets)	0.43	0.33	0.07	0.16	0.77	0.62
WTBlade	ViT-B 100%	0.58	0.27	0.06	0.09	0.85	0.81

Table 14. Box extent against the annotator ceiling on WTBD (same-class IoU of a label with its copy, or of the best prediction with conf ≥ 0.1, per test box). RF-DETR-M's gap is in finding defects; once found, its boxes match labels as well as a second annotator's do. The YOLO-family runs were extent-limited.

source	boxes	IoU ≥ 0.5	of those, IoU ≥ 0.75	median IoU of those	IoU ≥ 0.75
annotators, all copies	2,965	0.87	0.70	0.83	0.61
annotators, test copies	392	0.91	0.75	0.84	0.68
RF-DETR-M 25% (4 subsets)	940	0.55	0.67	0.83	0.37
ViT-B 25% (4 subsets)	940	0.41	0.45	0.73	0.19
BladeYOLO-v2 100%	235	0.63	0.47	0.74	0.29

- **RF-DETR at 25% labels proposes the right box for most defects.** Candidate recall is 0.92 on WTBD (above BladeYOLO-v2 at 100%, 0.80) and 0.83 on WTBlade. Its deficit is scoring: 37% of WTBD test defects have a correct box scored below 0.1.
- **On WTBD the rare classes are confused**, with crack missed or mislabelled and thunderstrike found under other classes (54% located as another class, in 4 subsets). On WTBlade, 43–49% of crack boxes lack a same-class box at IoU ≥ 0.5 at every budget (RF-DETR 8%: 0.47; 25%: 0.49; ViT-B 100%: 0.43). So this is not a label-budget problem. Misses concentrate on small boxes (1.00 below 16 px, 0.65 below 32 px, 0.22 above 64 px), and among near-misses the largest share is shifted boxes (44%), not split ones.
- **Not resolution, on WTBD.** Median defect short sides at 640 px are 62–148 px by class.

8 Leave-one-dataset-out: data volume does not replace target labels

If label count were the bottleneck, pooling every other blade dataset should help. G28 tests this directly. It was pre-registered before any pool was built or run started.

8.1 Design

For each target T, two class-agnostic RF-DETR-M arms (640 px, cosine, seed 0) are scored on T's test set:

- **ID-T:** T's group train (WTBD 100 epochs, WTBlade 60).

- **X-T:** every other source, twin-free against *all* of T, 12 epochs. Selection uses a held-out 5% of X-T's own images [32]; T is never seen, not even for selection.

Every image of T is an evaluation key for the twin rule: same file, DINOv2 ≥ 0.90, ORB ≥ 25 inliers, pixel correlation ≥ 0.97 over 8 orientations, and the closure of all of these. GAN images and box-less images are dropped. The rule deliberately sits below the audit's twin thresholds (0.93 and 0.99): earlier versions of this rule, on the G27 pool, kept 830 pixel twins and then neighbouring video frames, and each failure was logged and fixed before any training. Fig. 21 shows the result: X-WTBD has 26,065 training images (55,845 boxes across its train and held-out images) and X-WTBlade 14,405 (27,844). Fig. 22 shows the closest pairs an independent audit could find; the highest pixel correlation and cosine to any target image are 0.964 / 0.827 and 0.970 / 0.900.

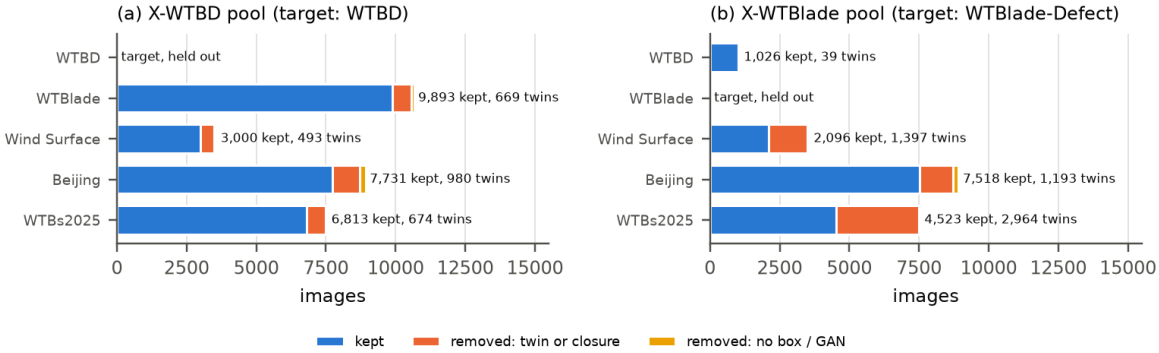


Figure 21. Composition of the G28 pools: images kept from each source and removed by the twin rule. For X-WTBlade, 2,964 WTBs2025 images are removed, since that dataset contains WTBlade-Defect's images.

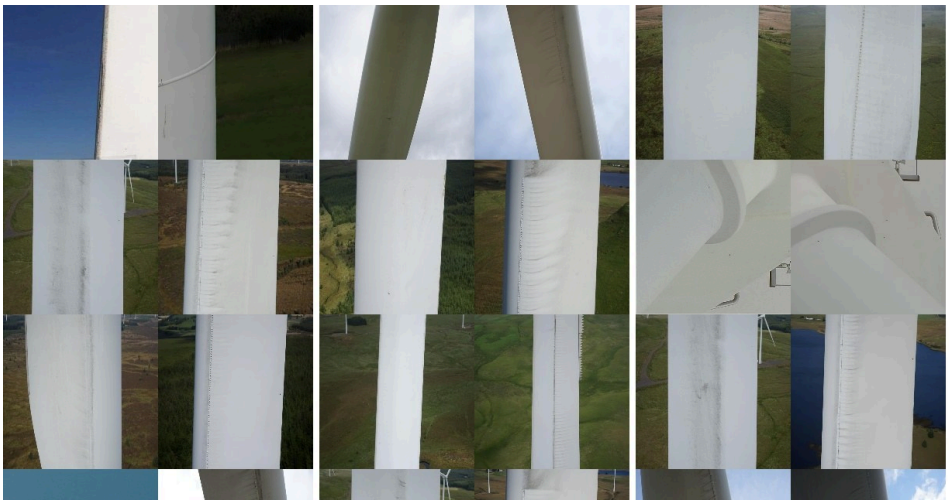


Figure 22. Independent audit of the X-WTBlade pool: the closest remaining pairs by pixel correlation between a kept pool image and a target image. By eye these are similar blades, not copies. The same physical defect from another flight cannot be excluded without capture metadata.

We define the gaps $G_{\mathrm{std}} = \mathrm{ID} - \mathrm{X}$ in standard AP50 and $G_{\mathrm{reg}} = \mathrm{ID} - \mathrm{X}$ in region AP, and the convention share $1 - G_{\mathrm{reg}}/G_{\mathrm{std}}$, the part of the gap that is about how boxes are drawn. The direction counted as alive only if, on *both* targets, X reached at least 0.80 × ID in region AP and the convention share was at least 0.5. Both bars were judgment calls fixed before any number existed.

8.2 Results

Table 15. G28, class-agnostic, test, scene bootstrap (2,000 replicates). *Ref* is G27 A0 (class-aware, 768 / 640 px), scored with classes merged; reported, not decisive. *Trivial* is one box over the whole image.

target	arm	train images	AP50	AP50-95	tolerant AP50	region AP	AR50
WTBD-group	ID	745	72.9 [57.6, 86.3]	41.9	84.6	68.9 [61.3, 75.4]	98.7
	X	26,065	**17.7** [11.9, 28.4]	6.0	24.2	43.6 [37.4, 52.3]	79.6
	ref	745	66.1	40.5	79.7	65.9	97.9
	trivial		0.0			14.0	
WTBlade-group	ID	2,136	51.7 [48.0, 58.8]	21.5	65.1	68.7 [62.6, 75.2]	90.5
	X	14,405	**2.4** [0.8, 3.5]	0.5	3.2	37.6 [32.5, 45.1]	38.2
	ref	2,136	50.1	21.3	61.2	68.0	92.2
	trivial		1.0			24.2	

Table 16. G28 decision. Bars: ratio ≥ 0.80 and share ≥ 0.5 on both targets. ID − X has P(> 0) = 1.000 on every metric of both targets.

target	X / ID region AP	convention share	G_{std} [95% CI]	G_{reg} [95% CI]	verdict
WTBD-group	0.63	0.54	+55.2 [+41.7, +66.4]	+25.3 [+14.4, +33.9]	dead
WTBlade-group	0.55	0.37	+49.3 [+45.0, +57.2]	+31.0 [+26.8, +34.6]	dead

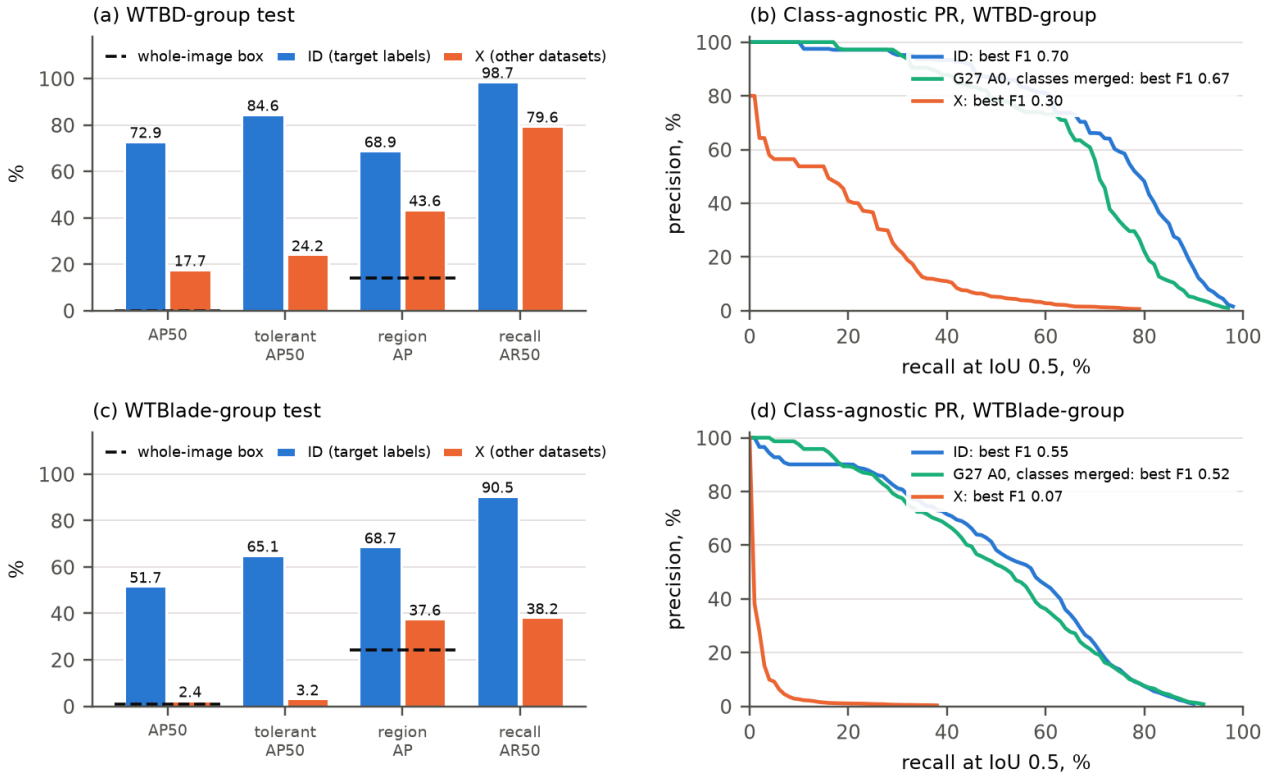


Figure 23. G28. (a, c) ID and X on target test; dashed lines mark a trivial whole-image detector. (b, d) Class-agnostic precision–recall at IoU 0.5.

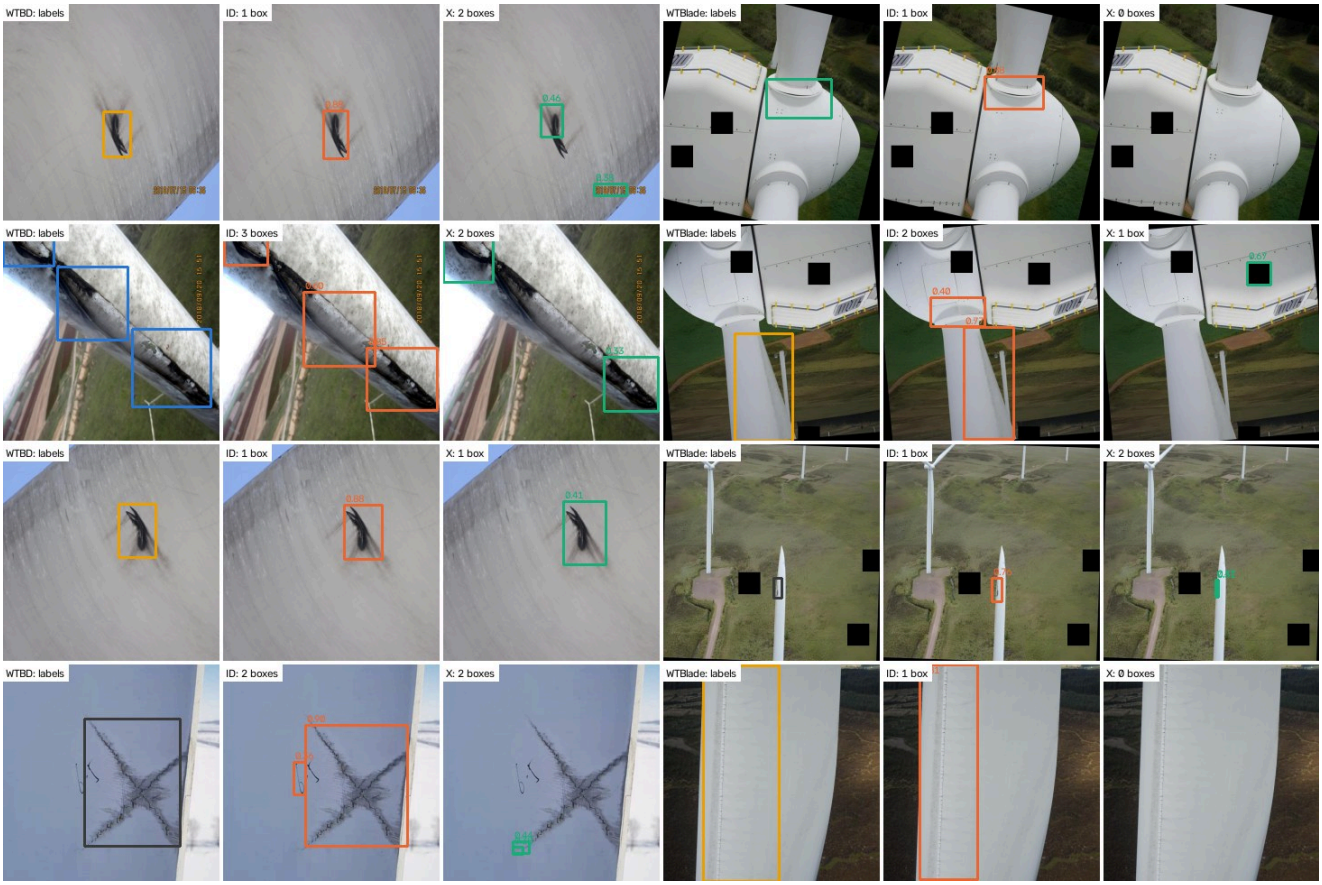


Figure 24. Labels, ID and X on random test images (fixed seed), boxes with confidence ≥ 0.3. Left three columns: WTBD-group. Right three: WTBlade-group. X often has the right area or a nearby distractor; it also returns nothing on images where ID is correct. Black squares in WTBlade-group are offline cutout from the Roboflow export, which the frames retain.

The direction fails on both targets. Table 15, Table 16 and Fig. 23 give the numbers, and Fig. 24 shows examples. With 35× more images from other sources, X still loses to 745 target images by 55 AP50 points on WTBD-group. The convention explains about half of the gap there and about a third on WTBlade-group. The remainder is a real failure to find the defect. Without the residual pixel-twin images of Fig. 5, the decision stays the same (ratio 0.64 and 0.56, share 0.55 and 0.39; ID AP50 71.9 and X 17.1 on WTBD-group).

The model did train. X-WTBD reached mAP50 72.2 on its own held-out images at its final epoch, and its region AP is above a trivial detector on both targets (43.6 against 14.0, 37.6 against 24.2). So it does carry information about where defects are. But it is not usable as a detector on an unseen dataset: X-WTBlade's AP50 of 2.4 is barely above boxing the whole image (1.0).

Why recall and AP disagree on WTBD. X finds 79.6% of WTBD labels at IoU 0.5 while scoring 17.7 AP50. Fig. 23 (b) shows why: true boxes rank among many false ones. At recall 10%, precision is 54% against 100% for ID; at recall 30% it is 23% against 95%. At its best F1 (0.30), X keeps about one box per image. On WTBlade-group, X's precision at recall 10% is only 3%, against 90% for ID. AR50 is generous because it counts every output, up to 300 per image, and some false positives are genuine damage of kinds T does not label, as noted in the pre-registration, so this test is conservative for X.

Earlier evidence agrees. Models trained on one benchmark transfer poorly to the other (Table 17), and an open-vocabulary foundation detector finds almost nothing without training. Pooling many sources raises the WTBD number from 5.2 to 17.7, but these are different models and scoring scopes, not a paired comparison, and on WTBlade-group pooling does not help (2.2–4.0 against 2.4).

Table 17. Generalisation outside one dataset, earlier experiments (BladeYOLO-v2 models; class-agnostic, damage classes only for transfer).

setting	result
WTBlade model on WTBD-group test	AP50 5.2 (in-domain 56.3, 58.0); difference −52.5 [−69.4, −32.9]
WTBD models on WTBlade-group test	AP50 4.0, 2.2 (in-domain 47.5); difference −43.9 [−49.6, −34.6]
Mapped classes across datasets	crack 0.0 both ways; thunderstrike / burn at most 7.3
OWLv2 [41], text queries, no training, WTBD-group	mAP50 3.3 (class-agnostic 5.7); trained on 25%: 25.0–25.7
OWLv2, few-shot prototypes from 25% labels	mAP50 0.6 (class-agnostic 0.5)

9 What did not work

The leak-free gap motivated many attempts. Reporting them matters, since a reader should know which common ideas have been tried here and how they fared.

9.1 G27: one recipe for all protocols

Diagnostics said that RF-DETR's extent is at the annotator ceiling, that candidate recall is high and that rare-class scoring is the loss, with post hoc rescoring closed. So G27 changed what the model learns from, with three additions on one pre-registered path:

- **CP:** copy-paste [33] of real boxes from the same training set to classes below the median count (Fig. 25).
- **D1:** in-domain self-supervised adaptation of RF-DETR's DINOv2 encoder by masked feature distillation [42], on a leak-checked pool of 22,318 images.
- **D2:** a multi-dataset supervised stage 1 on the same pool, with partitioned class slots [31].

A-1 selected the recipe R on validation, seed 0: the best mean among arms beating A0 on both datasets. A-2 is the primary test claim on seeds 0 and 1: $R - A0 > 0$ on each dataset, with pooled $P(> 0) \geq 0.95$.

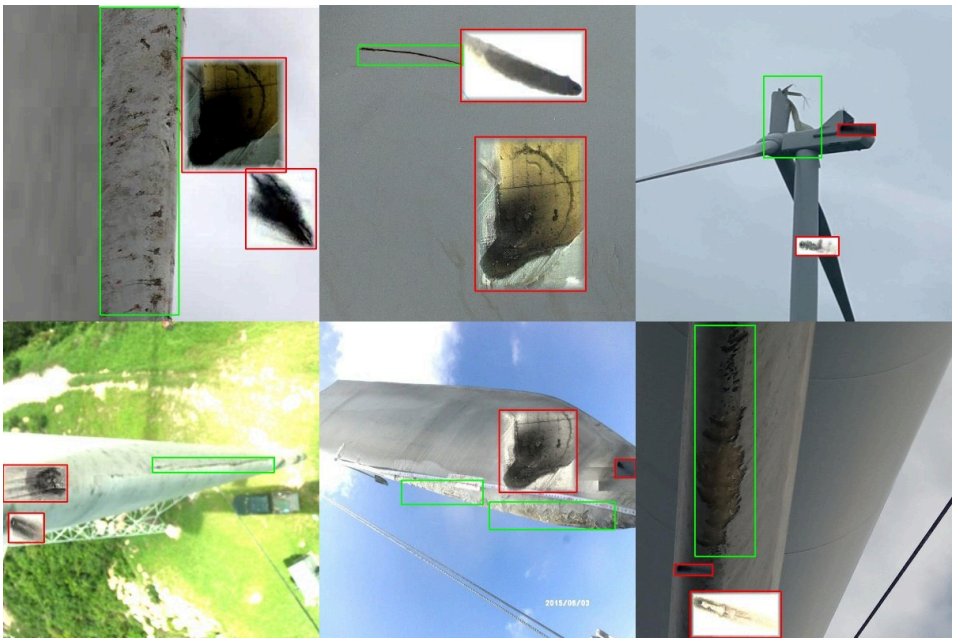


Figure 25. Visual check of copy-paste, as made before training. Some pastes land off the blade where SAM's [43] largest mask includes sky, and box-level pastes carry their source background. These cases were kept, since the A1 − A0 comparison measures the recipe as it ran.

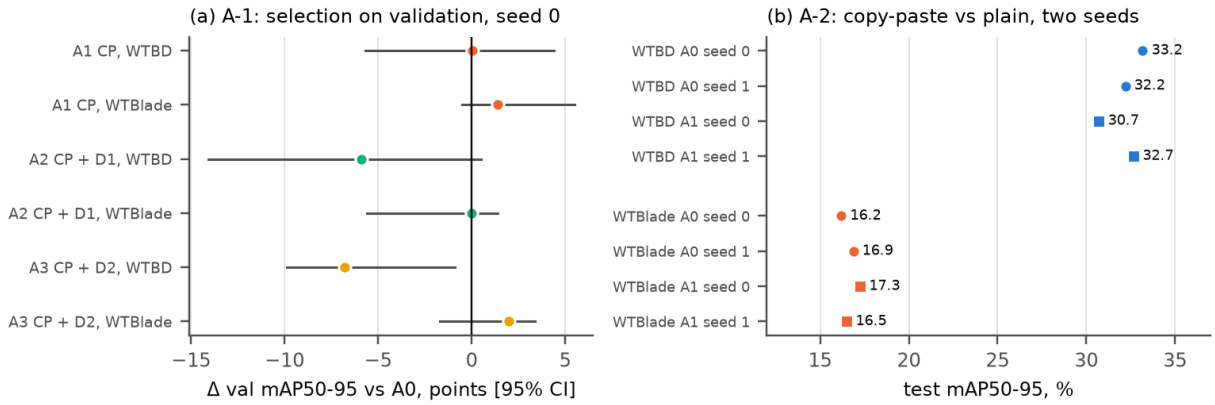


Figure 26. G27. (a) A-1, validation deltas against A0, seed 0. (b) A-2, test by seed: the seed swing is as large as the effect.

Table 18. G27 A-2, R = A1 (copy-paste), R − A0, test mAP50-95, mean of seeds 0 and 1. The primary claim fails.

	R − A0	95% CI	P(> 0)
WTBD-group	−0.99	[−8.6, +1.2]	0.193
WTBlade-group	+0.35	[−0.5, +1.6]	0.812
mean over datasets	−0.32	[−4.1, +0.9]	0.290

As pre-registered, we also score without the residual pixel-twin images: R − A0 is −1.01 on WTBD-group and +0.40 on WTBlade-group, so the conclusion is unchanged.

R = A1 was chosen by +0.04 points on WTBD-group validation, smaller than A0's own seed-to-seed difference on validation. D1 and D2 were rejected on A-1 (−5.9 and −6.8 points against A1 on WTBD val), and a post hoc check found D1 halved the encoder's last-stage effective rank (257 to 135), partial dimensional collapse. A seed swing of up to 2 points on these test sets (A1 on WTBD: 30.7, 32.7) sets the smallest effect that can be claimed.

Table 19 gives the validation scores that made the selection, with the test scores beside them.

Table 19. All G27 arms, mAP50-95 from the G27 scorer. Validation decided R; test was not used for selection.

arm	WTBD-group val, seed 0	WTBD-group test, seeds 0 / 1	WTBlade-group val, seed 0	WTBlade-group test, seeds 0 / 1
A0 plain	40.11	33.2 / 32.2	15.68	16.2 / 16.9
A1 + CP	40.15	30.7 / 32.7	17.10	17.3 / 16.5
A2 + CP + D1	34.2	27.4 / –	15.6	16.1 / –
A3 + CP + D2	33.4	33.2 / –	17.7	16.6 / –

9.2 Semi-supervised learning in detail

Label budgets suggested semi-supervision, and it took three rounds to rule out (Table 20). Seed noise alone is ±2.6 mAP50 at 25%, as wide as every effect measured. The class-balanced variant is significantly worse, partly through a confirmation-bias runaway: one seed produced up to about 2,300 pseudo-boxes per epoch from 24 images. Measured against hidden labels, even a good teacher's pseudo-labels are 66% correct at threshold 0.5 and 40% at 0.25.

Table 20. Teacher–student rounds at 25% of WTBD-group [44]. Top: one seed per arm, COCO-style evaluator. Bottom: three seeds per arm.

arm	test mAP50 / mAP50-95	val best mAP50 / mAP50-95	pseudo-labels
supervised	23.2 / 8.2	44.4 / 22.6	–
synthetic views, unlabelled pool	22.4 / 7.8	44.8 / 22.0	2,748
real views, unlabelled pool	19.2 / 7.7	38.2 / 18.4	961
in-domain (the rest of train, labels hidden)	20.9 / 8.0	45.7 / 22.7	1,294
supervised, 3 seeds	21.3 ± 2.6 / 7.9 ± 1.0		–
class-balanced in-domain, 3 seeds	16.3 ± 2.0 / 6.6 ± 0.6		5,957, 28,742, 4,372

9.3 The ledger

Table 21 lists the method hypotheses we ran or formally gated, with their results. Fig. 27 plots those with a test effect in points. Some effects are real but narrow (copy-paste on rare classes in a low-label setting), and few survive a second dataset or a replication.

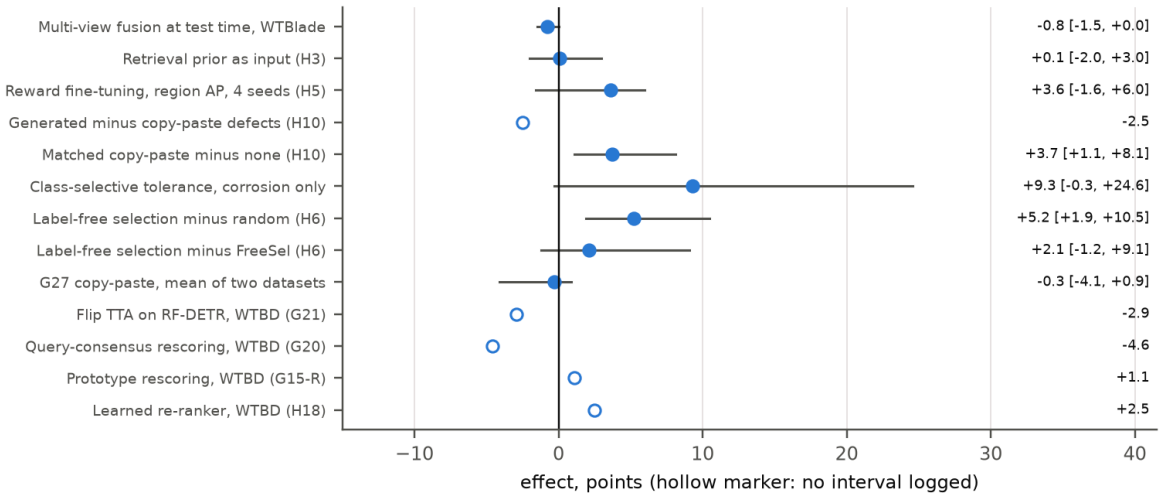


Figure 27. Effects of method tests on leak-free test sets, in points of mAP50-95 unless the ledger says otherwise. WTBD-group unless labelled. Hollow markers: no interval was logged in that form.

Table 21. Ledger of method tests and diagnostics. WTBD-group, test mAP50-95 points unless noted. Pre-registered from late October; earlier rounds were decided in advance but not formally committed.

test	idea	result	verdict
Phases 22–24	Teacher–student with synthetic or real views, in-domain, class-balanced [44]; 3 seeds	balanced in-domain 6.6 vs supervised 7.9; mAP50 −4.2 [−12.1, −0.3]; confirmation-bias runaway on one seed	fails
Phase 24	SAM 2.1 box refinement [43]	+0.3 to +0.5 at 25%, −0.2 at 100%	fails
Phases 18, 26	Split-tolerant training; class-selective version, 2 seeds	blanket not an improvement; corrosion +9.3 [−0.3, +24.6], mean −0.4	targeted only
H1	Multi-view fusion at test time	not testable on WTBD (26 images with a second view); WTBlade-group −0.8 [−1.5, 0.0]	fails
H2	Normal-blade anomaly prior	most anomalous patches not on defects; box recall below chance	fails
H3	Retrieval prior as a detector input	+0.1 [−2.0, +3.0]	fails
H4	Test-time adaptation across real views	gate failed: real views do not make better detections	not run
H5	Reward fine-tuning, granularity-invariant reward; 4 seeds + WTBlade	region AP +3.6 [−1.6, +6.0], P 0.896; WTBlade −0.6	fails on replication
H6	Label-free scene selection by patch coverage	+5.2 [+1.9, +10.5] over random; +2.1 [−1.2, +9.1] over FreeSel	no gain over prior art
H7	Label-free property predicting subset quality	no property survives correction	fails
H8, H9 (G8, G9)	Feature upsampling; high-frequency cues	−5.9; −0.1 (mean AP)	fail
H10	Generated rare-class defects vs matched copy-paste	synth − paste −2.5; paste − none +3.7 [+1.1, +8.1]	generation fails; copy-paste helps here
H12	DINOv3 reduces subset luck	it does not; draw quality carries over (ρ 0.71)	fails
G15, G15-R	Prototype rescoring	WTBD +1.1; WTBlade −0.7	fails bar; prior art
G16, G16-R	Instance-level proximity law; explains subset luck?	law holds at instance level; does not explain subset luck; on RF-DETR it holds for confidence, not for detection	partial
H16b	Deployment-targeted scene selection	gate failed	not run
G17	Clicks as pseudo-boxes (bar: 60% at IoU ≥ 0.5)	42.0%	fails
H18, -R, -W	Learned coverage-aware re-ranker on RF-DETR	WTBD +2.5, replicated on new models; WTBlade +0.2	WTBD only
H19-W	Rescoring at the low-count boundary, WTBlade 8%	−0.2 [−1.3, +0.7], P 0.286	fails
G20	Query-consensus rescoring	−4.6 WTBD; −2.1 WTBlade	fails
G21	Flip agreement; plain flip TTA on RF-DETR	−2.9 WTBD; −0.6 WTBlade	fails; TTA hurts this DETR
G22, -O	Fine-tuning moves proximity from recall to ranking	controlled confirmation on WTBD; not on WTBlade; holds only for generic-feature proximity	does not replicate
H20	Subset luck as annotation-convention match	no relation	fails
G24	Input resolution 768 vs 576	WTBD +1.0 (P 0.958); WTBlade −0.1	dataset-specific
V1	RF-DETR replaces the frozen-feature detector at 25%	+7.3 [+2.3, +11.9]; +8.3 [+4.5, +15.0]	passes

test	idea	result	verdict
Box extent; crop re-classifier	Better box refinement; re-classifying RF-DETR candidates	prior-art and ceiling checks show little room	discarded before any run
G25, G26	Official-protocol benchmark; tuned recipe	Table 2, Table 5	reported
G27	CP, D1, D2 recipe	A-2 −0.32, P 0.290	fails
G28	Leave-one-dataset-out	dead on both targets	fails

10 A cautionary history: how our own numbers were wrong

We report this because the same traps face anyone working on these datasets. Fig. 28 gives the sequence, and Table 22 the original runs.

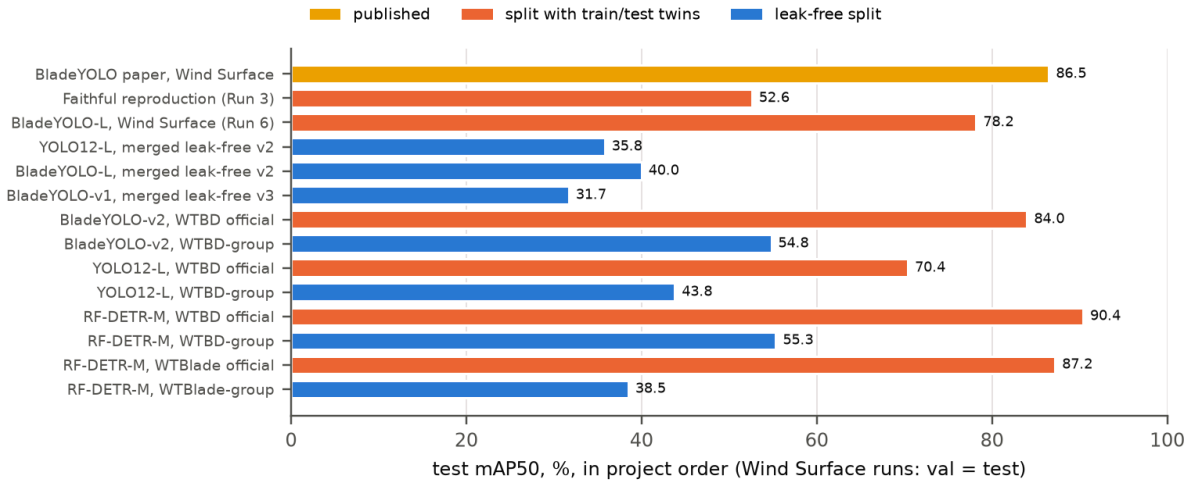


Figure 28. Reported mAP50 at each stage of the project, in order. Every class-aware score above about 60 in this figure was measured on a split with train/test twins, including our own. Datasets differ between rows, so rows are not comparable as models; the point is the pattern.

Table 22. Early runs on the original Wind Surface split (2,856 train / 714 val). Later analysis invalidated the conclusions drawn from them.

run	architecture	epochs	mAP50	mAP50-95	outcome
1–2	paper-faithful: DINOv3 + Style Injection + WTConv + Cross-Mamba	crash, ~200			bf16, unfreezing and DDP failures; NaN from FFT gradients
3	paper-faithful, FFT detached, SGD	~283	~52.6	~28	stagnated; closest reproduction
4	BladeYOLO-S (8 M parameters)	400	64.3	37.3	apparent ceiling at about 64
6	BladeYOLO-L (57.2 M), AdamW	400	78.2	46.9	looked like a breakthrough
7	Run 6 + “Cross-Scale Vision Mamba” + WIoU v3	400	77.0	47.8	“+0.9 from WIoU”

The gap of 33.9 / 29.4 points between Run 3 and the published 86.5 / 57.4 is far beyond run-to-run variance. These runs used a weakened feature extractor (below) and the leaky split, so they support “not reproducible from the published information”, and not “the published modules do not work”. The paper reports 3,790 images while the public copy has 3,570, and gives no split seed, image list or code.

Then we audited our own code. Run 7 was not the model we described:

- **WIoU never ran**, because the patch reached the wrong namespace, so Run 7 trained with CIoU. Its “+0.9 mAP50-95 from WIoU” was invalid. The formula also had three errors.
- **The “Mamba” block did not perform a selective scan.** It was replaced by a four-direction implementation that matches a reference recurrence to 1.8e-14.
- **The DINOv3 extractor skipped RoPE and fed unnormalised pixels**, so every run before 9 October used weakened features.
- **The wavelet branch discarded high-frequency detail.**

Then we found the leak. Removing it took the numbers from about 78 to about 36–40 on a merged, deduplicated dataset (v2, 2,606 unique images from 11,681), and a larger taxonomy-mapped v3 (5,777 images) scored only 26–32. An exhaustive label review of 763 candidates was worth only +3 mAP50 points. v3’s limit was structural: its classes rest on

few independent scenes, and one had a single training group. Only then did the audit lead us to WTBD alone, with its native classes.

Table 23 in the appendix lists all conclusions we later corrected. Several were our own favourite explanations.

11 Recommendations

1. **Split by scene, and deduplicate before augmenting.** Never augment before a split. Group by source photo, video and tile, and add a pixel test: retrieval and keypoints alone missed photometric copies in our own splits.
2. **Publish image lists**, not just ratios. The BladeYOLO 0.87-ratio question cannot be settled without one.
3. **Report a leaked / no-twin breakdown** for any dataset with known copies. It takes minutes with cached predictions, and our tools are public.
4. **Do not tune on a leaky test set.** On these splits tuning measurably moved scores towards memorisation. Tune on a scene-separated validation set.
5. **Use scene-level intervals.** A 161-image test set with 58 scenes, and some classes in fewer than 3 scenes, cannot support per-class claims.
6. **Report several labelled subsets for low-label claims.** A factor of two between random draws is larger than any model effect we saw.
7. **Report seeds.** On these test sets, seed swings of up to 2 mAP50-95 points match the size of most claimed gains.
8. **Calibrate on unseen scenes.** A recall guarantee calibrated on seen scenes does not hold in the field.
9. **Report a granularity-invariant score beside box AP** for amorphous defects, and measure label self-consistency where copies exist.
10. **Test every method on a second dataset.** Several of our effects held on WTBD and vanished on WTBlade, and vice versa.
11. **Check licences at the source**, since export pages can misstate them.

12 Limitations

- **Single wind farm per dataset; small test sets** (58 and 177 test scenes). Our leak-free numbers carry wide intervals.
- **Official-protocol runs are single seed**, and their intervals are image-level.
- **The tuning analysis is post hoc and covers one step**, with epochs confounded on two of three protocols.
- **Wind Surface's "no twin" subset is not leak-free** (it shares drone photos with train), and WTBD's has too few boxes of three classes.
- **WTBlade-group frames retain Roboflow's offline augmentation.** We keep one copy per source, so some test frames are rotated or have black cutouts (Fig. 24). This is a remaining difference from clean imagery, not a leak.
- **YOLO12-L on WTBlade-group may be under-trained:** 50 epochs, the budget planned for BladeYOLO-v2 ViT-B on this set, with its best validation at epoch 43 after a nearly linear rise. Its 13.3 may understate it.
- **Our leak-free splits have a small residual leak** (§ 4, Fig. 5): under a point on WTBD-group, negligible on WTBlade-group. G28's ID arm benefits from it slightly.
- **G27 A-2 uses two seeds; G28 uses one.** G28 X arms train for 12 epochs, as G27's stage 1, at 640 px. X-WTBlade loses the largest source, and its pool is 45% smaller than X-WTBD's, which may contribute to the difference between targets.
- **Hardware differs between seeds.** G27 seed 0 ran on Kaggle (fp16) and seed 1 on an A100 (bf16). Arms are paired within a seed, but the seed effect includes hardware. An earlier A0 seed-1 run on Kaggle scored 31.2 on WTBD-group test against 32.2 on the A100.
- **Region AP depends on defect size.** A trivial whole-image detector scores 14–24 here, so raw region AP must be read against that floor.
- **Claims about modules are for our components.** BladeYOLO's own modules were not rerun on a leak-free split after the code audit.
- **Deviations from pre-registration.** The G26 rule formally selected RF-DETR-L (validation 63.4 against 62.0), which lost on test (−1.7), and we did not run L on the other protocols; this cannot inflate a reported number, but it is a deviation. G27's phase B on the official protocols was not run, since the recipe failed A-2.

- **The literature check is not exhaustive**, and "new for this domain" means that we did not find an earlier audit.

13 Data and code availability

Code and records. All code, tests, split files, per-image flags and the full research log (`research_log.md`) are at https://github.com/bhavya-goel-11/BladeYOLO. To regenerate this paper's numbers and figures, run `tools/official_leak_eval.py`, `tools/g27_score.py`, `tools/g28_score.py`, `tools/paper_evidence.py`, `tools/paper_figures.py`, `tools/paper_figures2.py` and `tools/build_paper.py`. The training queue is `tools/sessions.sh`.

Data. The datasets, leak-free splits and pools are assembled as `blade-defect-benchmark`, with per-folder licences in its `LICENSES.md`. WTBD is CC BY 4.0. WTBlade-Defect contains DTU NordTank images [5] under CC BY-NC, despite its Roboflow page, and derived folders inherit this, so they are for non-commercial research only. The Wind Surface licence is unclear from its source. Images in our figures are shown for scientific comment; sources are cited above.

Foundation models. We acknowledge DINOv3 [21], used under the DINOv3 License, and DINOv2 [19] through RF-DETR [18]. Trained checkpoints containing DINOv3 weights must ship with that licence.

Appendix

A. Corrections to our own conclusions

Table 23. Conclusions we later corrected. Kept here because each was plausible at the time.

earlier conclusion	status	why
Runs 6–7 reach ~78 / ~47 on Wind Surface	invalid	75% of validation had train twins (first count)
WIoU v3 gave +0.9 mAP50-95	invalid	never active; Run 7 trained with CIoU
Mamba added global context	invalid	the block did not scan
SGD → AdamW was the breakthrough	confounded	changed together with an 8 M → 57 M scale-up
The remaining gap is locality bias	superseded	the gap was leakage and data structure
v3 is better because it is larger	superseded	scenes matter, not files (Wind Surface 81; WTBD 550)
Mislabelling explains poor v3 scores	rejected	the review was worth only +3 mAP50 points
BladeYOLO-v2's win shows the architecture generalises	qualified	it rests mainly on thunderstrike (3 test scenes)
The advantage comes from the architecture	rejected	the frozen DINOv3 features carry it all
Honest calibration costs 2.4 vs 1.0 false alarms per image	corrected	that column counted all predictions; real false alarms 0.2 vs 0.1
WTBD copies agree on 86.0%	refined	rotated copies inflate "loose"; axis-aligned 87.3%
Part of the inflation is memorised box splitting	not supported	tolerant gain is not smaller on images with a twin
At 25%, most lost defects are found but boxed differently	refined	near-misses (34–36%) and misses (19–27%) are of similar size
Split tolerance helps corrosion	replicated, not significant	+9.3 [−0.3, +24.6]; aggregate unchanged
The DINOv3 advantage is a low-data advantage	corrected	absolute gain does not shrink with data
The gain grows with frozen backbone size	partly supported	ViT-B beats ViT-S; ViT-L does not beat ViT-B
Teacher–student gives a small consistent gain	qualified	seed noise was as large; balanced variant significantly worse
Leak-free mAP50-95 / mAP50 ratios are 0.42–0.47	corrected	full-label RF-DETR on WTBD-group reaches 0.60

B. Engineering problems that affected results

Table 24. Defects we found and fixed. Several of them silently changed what was being measured. Mixed-precision overflow was the most frequent.

problem	root cause	fix
WIoU never active	patched the wrong namespace	patch the loss namespace
NaN in selective scan (epoch 9)	fp16 overflow before LayerNorm	fp32 LayerNorm
NaN at epoch 4	poisoned WIoU running mean; aggressive warm-up	finite-only mean; lr 0.001, bias warm-up 0
NaN at epoch 17	fp16 overflow from a conv into BN	fp32 blocks for refinement
NaN at epoch 99, validation at 0	NaN batch poisons BN statistics and EMA	per-batch BN repair and step skip, reported
NaN late in training (epochs 99–149)	YOLOv12 area attention forms scores in fp16 (~4e5, above the fp16 maximum)	scores and softmax in fp32
Test metrics silently missing	`deform_conv2d` segfault on NaN offsets from the FLOPs probe's uninitialised image	finite, clamped offsets; test in a fresh process
Silent resume from an old checkpoint	training script reused any `last.pt`	explicit `--resume`
RF-DETR learning rate never decayed	default step at epoch 100 never fires	explicit cosine schedule (G26)
Scorer read a stale result after a time-out	an earlier output file had the same name	rerun with no limit; check file freshness

C. Notation and experiment names

Names follow the research log: H*n* and G*n* are hypotheses and gates in order of registration. A0–A3 are the G27 arms: A0 plain recipe, A1 + copy-paste, A2 + copy-paste + D1, A3 + copy-paste + D2. ID and X are the G28 arms. "Points" are percentage points of AP. WTBD-official is WTBD's own split.

D. Per-class audit

Table 25. Content audit by native class (`runs/analysis/datasets/report.md`). *Learnable* boxes are those kept after deduplication. *Purity* is mutual crop retrieval within the class. Median box size is in pixels at 640. The check export of Beijing is omitted because it is the same data.

dataset	class	raw images	raw boxes	unique images	scenes	learnable boxes	median box px	purity
Wind Surface	corrosion	660	2116	90	10	294	87.7	0.894
Wind Surface	dirt	1044	1067	487	14	488	257.3	0.923
Wind Surface	hide_craze	1305	2321	272	50	491	53.6	0.719
Wind Surface	surface_eye	108	412	9	1	40	85.5	0.0
Wind Surface	thunderstrike	364	478	70	10	89	55.5	0.12
WTBlade	burn	736	1019	116	104	166	56.9	0.65
WTBlade	crack	1409	2525	177	112	335	63.0	0.573
WTBlade	deformity	1801	2291	282	231	383	130.7	0.802
WTBlade	dirt	5148	8875	585	425	928	255.2	0.683
WTBlade	oil	539	958	97	82	167	105.6	0.428
WTBlade	peeling	2307	5715	172	80	395	96.5	0.583
WTBlade	rust	2043	4188	157	123	258	95.2	0.354
Beijing	corrosion	789	2565	90	10	292	86.9	0.666
Beijing	craze	57	69	9	8	12	158.3	0.108
Beijing	hide_craze	1187	2137	283	52	499	54.9	0.545
Beijing	surface_eye	3316	5489	900	17	1467	227.3	0.888
Beijing	surface_injure	1662	3285	290	45	564	85.3	0.635
Beijing	surface_oil	1588	2242	417	27	574	249.9	0.473
Beijing	thunderstrike	270	353	69	10	87	52.2	0.153
WTBs2025	oil leakage	512	572	28	28	35	33.9	0.486
WTBs2025	paint cracks	458	514	398	351	235	17.8	0.666
WTBs2025	localized damage	1027	1254	773	655	86	6.5	0.256
WTBs2025	lightning strikes	285	354	89	43	96	43.1	0.56
WTBs2025	surface stains	263	294	226	211	142	23.5	0.182
WTBs2025	erosion	2798	4517	1132	993	1174	21.4	0.675
WTBs2025	coating detachment	1085	1906	703	566	526	15.3	0.484
WTBs2025	protective film damage	546	632	393	350	195	14.2	0.639

dataset	class	raw images	raw boxes	unique images	scenes	learnable boxes	median box px	purity
WTBs2025	pinholes	522	774	377	311	45	2.2	0.136
WTBD	craze	198	257	94	86	119	206.5	0.555
WTBD	corrosion	172	257	105	100	149	194.3	0.723
WTBD	surface_injure	264	412	164	155	247	103.3	0.723
WTBD	thunderstrike	89	92	50	47	52	138.0	0.565
WTBD	crack	177	224	47	37	55	171.5	0.842
WTBD	hide_craze	249	342	182	171	244	111.9	0.785

E. Run registry

Table 26. Runs of the final phase (G27, G28, YOLO12-L). Val best is the best EMA epoch (Ultralytics best epoch for YOLO12-L). Test values are from Table 7 and Table 15.

run	data	epochs (best)	val best mAP50 / mAP50-95	test mAP50 / mAP50-95
`rfdetrm768cos_wtbd_group_full`	WTBD-group, A0, s0	100 (20)	64.8 / 40.4	55.3 / 33.2
`rfdetrm768cos_wtbd_group_full_s1`	WTBD-group, A0, s1	100 (32)	67.1 / 40.8	54.8 / 32.2
`rfdetrm768cos_cp_wtbd_group_full`	WTBD-group, A1, s0	100 (9)	63.9 / 40.1	51.5 / 30.7
`rfdetrm768cos_cp_wtbd_group_full_s1`	WTBD-group, A1, s1	100 (31)	70.7 / 41.8	58.5 / 32.7
`rfdetrm768cos_cp_d1_wtbd_group_full`	WTBD-group, A2	100 (63)	59.6 / 34.4	49.6 / 27.4
`rfdetrm768cos_cp_d2_wtbd_group_full`	WTBD-group, A3	100 (40)	56.8 / 33.8	55.3 / 33.2
`rfdetrm640cos_wtblade_group_full`	WTBlade-group, A0, s0	60 (11)	38.4 / 15.7	38.5 / 16.2
`rfdetrm640cos_wtblade_group_full_s1`	WTBlade-group, A0, s1	60 (17)	41.1 / 16.2	41.0 / 16.9
`rfdetrm640cos_cp_wtblade_group_full`	WTBlade-group, A1, s0	60 (13)	41.6 / 17.1	41.8 / 17.3
`rfdetrm640cos_cp_wtblade_group_full_s1`	WTBlade-group, A1, s1	60 (16)	41.3 / 16.9	39.3 / 16.5
`rfdetrm640cos_cp_d1_wtblade_group_full`	WTBlade-group, A2	60 (16)	39.2 / 15.7	40.1 / 16.1
`rfdetrm640cos_cp_d2_wtblade_group_full`	WTBlade-group, A3	60 (8)	44.2 / 17.8	41.7 / 16.6
`rfdetrm640ag_id_wtbd_group`	ID-WTBD, class-agnostic	100 (23)	80.7 / 48.3	72.9 / 41.9
`rfdetrm640ag_id_wtblade_group`	ID-WTBlade, class-agnostic	60 (26)	49.5 / 20.5	51.7 / 21.5
`rfdetrm640ag_x_wtbd_group`	X-WTBD, class-agnostic	12 (12)	72.2 / 38.6 (own val)	17.7 / 6.0
`rfdetrm640ag_x_wtblade_group`	X-WTBlade, class-agnostic	12 (12)	62.1 / 33.6 (own val)	2.4 / 0.5
`yolo12l_wtblade_group`	WTBlade-group	50 (43)	33.6 / 12.5	32.2 / 13.3

Table 27. Registry of earlier runs, as kept in the research log (§27 there; section references inside this table are to the log). Scores are fractions, not percent.

Run folder	Data	Model	Epochs	Val best (mAP50 / 50-95)	Test (mAP50 / 50-95)	Notes
Runs 1–5	WS original	Paper-faithful / BladeYOLO-S	—	see §2	—	Logs on Kaggle only
`bladeyolo_l_sota` (Run 6)	WS original (**leaky**)	BladeYOLO-L	400	0.781 / 0.469	—	Inflated
`bladeyolo_l_sota-2` (Run 7)	WS original (**leaky**)	BladeYOLO-L + “Mamba” + (inactive) WIoU	400	0.769 / 0.478	—	Inflated
`yolo12l_baseline`	v2	YOLO12-L	266	0.290 / 0.122	0.358 / 0.149	lr0 0.002
`bladeyolo_l`	v2	BladeYOLO-L (repaired)	167 (stopped)	0.344 / 0.123	0.400 / 0.146 (epoch 124)	
`yolo12l_v3`	v3	YOLO12-L	150	0.290 / 0.141	0.263 / 0.109	Old harmful recipe
`bladeyolo_v3`	v3	BladeYOLO-L v1	150	0.356 / 0.156	0.317 / 0.124	
`bladeyolo2_v3`	v3	BladeYOLO-v2	100	0.369 / 0.172	0.296 / 0.123 (evaluator)	Test lost on Kaggle (segfault)
WBF ensemble	v3 / v3.2	3 models	—	—	0.339 / 0.143 (v3); ~0.411 / 0.168 (merged)	Fusion IoU chosen on val
`bladeyolo2_wtbd_group` (attempt 1)	WTBD-group	BladeYOLO-v2	99 of 150	0.649 / 0.379	—	NaN at epoch 99; folder lost

Run folder	Data	Model	Epochs	Val best (mAP50 / 50-95)	Test (mAP50 / 50-95)	Notes
bladeyolo2_wtbd_group	WTBD-group (leak-free)	BladeYOLO-v2	150	0.650 / 0.368 (epoch 124)	**0.548 / 0.245**; TTA 0.574 / 0.279	Test run locally
bladeyolo2_wtbd_official	WTBD official (**leaky**)	BladeYOLO-v2	120	0.845 / 0.475 (epoch 105)	**0.840 / 0.493**; TTA 0.830 / 0.517	TTA run locally
bladeyolo2_wtbd_group_tolerant	WTBD-group (leak-free)	BladeYOLO-v2, split-tolerant targets	150	0.609 / 0.352 (epoch 83)	**0.523 / 0.249**; TTA 0.552 / 0.282	5 NaN batches dropped (attention overflow, since fixed)
bladeyolo2_wtbd_group_f25	WTBD-group, 25% of train scenes	BladeYOLO-v2	150	0.444 / 0.226 (epoch 126)	**0.232 / 0.082**; TTA 0.269 / 0.105	Learning curve
bladeyolo2_wtbd_group_f50	WTBD-group, 50% of train scenes	BladeYOLO-v2	150	0.472 / 0.263 (epoch 140)	**0.413 / 0.183**; TTA 0.473 / 0.211	Learning curve
yolo12l_wtbd_group_f25	WTBD-group, 25% of train scenes	YOLO12-L	150	0.233 / 0.139 (epoch 150)	**0.106 / 0.037**; TTA 0.144 / 0.059	Learning curve
yolo12l_wtbd_group_f50	WTBD-group, 50% of train scenes	YOLO12-L	150	0.279 / 0.167 (epoch 120)	**0.244 / 0.105**; TTA 0.259 / 0.125	Learning curve
bladeyolo2_wtbd_group_f25_sslsyn	WTBD-group 25% + view pool	BladeYOLO-v2, synthetic-view teacher–student	150	0.427 / 0.203 (epoch 142)	**0.221 / 0.071**; TTA 0.233 / 0.089	Phase 22
bladeyolo2_wtbd_group_f25_sslreal	WTBD-group 25% + view pool	BladeYOLO-v2, real-view teacher–student	150	0.450 / 0.210 (epoch 107)	**0.213 / 0.085**; TTA 0.268 / 0.107	Phase 22; teacher collapsed
bladeyolo2_wtbd_group_f25_sslsyn2	WTBD-group 25% + view pool	BladeYOLO-v2, synthetic views (fixed)	150	0.448 / 0.220 (epoch 122)	**0.224 / 0.078**; TTA 0.259 / 0.106	Phase 23
bladeyolo2_wtbd_group_f25_sslreal2	WTBD-group 25% + view pool	BladeYOLO-v2, real views (fixed)	150	0.382 / 0.184 (epoch 127)	**0.192 / 0.077**; TTA 0.248 / 0.101	Phase 23
bladeyolo2_wtbd_group_f25_sslin	WTBD-group 25% + other 75% unlabelled	BladeYOLO-v2, in-domain teacher–student	150	0.457 / 0.227 (epoch 131)	**0.209 / 0.080**; TTA 0.245 / 0.101	Phase 23
bladeyolo2_wtbd_group_f25_s1	WTBD-group, 25% of train scenes	BladeYOLO-v2, seed 1	150	0.411 / 0.199 (epoch 129)	**0.224 / 0.087**; TTA 0.242 / 0.105	Phase 24
bladeyolo2_wtbd_group_f25_s2	WTBD-group, 25% of train scenes	BladeYOLO-v2, seed 2	150	0.435 / 0.203 (epoch 143)	**0.184 / 0.068**; TTA 0.229 / 0.096	Phase 24
bladeyolo2_wtbd_group_f25_sslinb_s0	WTBD-group 25% + other 75% unlabelled	BladeYOLO-v2, in-domain, class-balanced, seed 0	150	0.344 / 0.154 (epoch 75)	**0.143 / 0.060**; TTA 0.183 / 0.079	Phase 24
bladeyolo2_wtbd_group_f25_sslinb_s1	WTBD-group 25% + other 75% unlabelled	same, seed 1	150	0.352 / 0.167 (epoch 92)	**0.163 / 0.065**; TTA 0.180 / 0.075	Phase 24; pseudo-label runaway, epochs 33–62
bladeyolo2_wtbd_group_f25_sslinb_s2	WTBD-group 25% + other 75% unlabelled	same, seed 2	150	0.445 / 0.205 (epoch 148)	**0.183 / 0.073**; TTA 0.213 / 0.095	Phase 24
bladeyolo2nodino_wtbd_group_f25_s0	WTBD-group, 25% of train scenes	BladeYOLO-v2 without DINOv3, seed 0	150	0.229 / 0.126 (epoch 106)	**0.114 / 0.044**; TTA 0.153 / 0.063	Phase 25
bladeyolo2nodino_wtbd_group_f25_s1	WTBD-group, 25% of train scenes	BladeYOLO-v2 without DINOv3, seed 1	150	0.186 / 0.105 (epoch 89)	**0.081 / 0.031**; TTA 0.126 / 0.053	Phase 25
bladeyolo2_wtbd_group_s1	WTBD-group (leak-free)	BladeYOLO-v2, seed 1	150	0.630 / 0.370 (epoch 128)	**0.554 / 0.245**; TTA 0.559 / 0.269	Phase 26; resumed at epoch 78
bladeyolo2_wtbd_group_tolsel_s0	WTBD-group (leak-free)	BladeYOLO-v2, tolerance for corrosion + hide_craze, seed 0	150	0.656 / 0.381 (epoch 140)	**0.567 / 0.245**; TTA 0.568 / 0.258	Phase 26
bladeyolo2_wtbd_group_tolsel_s1	WTBD-group (leak-free)	same, seed 1	150	0.704 / 0.380 (epoch 119)	**0.561 / 0.245**; TTA 0.555 / 0.264	Phase 26
bladeyolo2b_wtbd_group_f25_s0	WTBD-group, 25% of train scenes	BladeYOLO-v2, frozen DINOv3 ViT-B/16, seed 0	150	0.383 / 0.176 (epoch 127)	**0.252 / 0.102**; TTA 0.313 / 0.127	Phase 27
bladeyolo2b_wtbd_group_f25_s1	WTBD-group, 25% of train scenes	same, seed 1	150	0.416 / 0.217 (epoch 146)	**0.283 / 0.116**; TTA 0.272 / 0.132	Phase 27
bladeyolo2l_wtbd_group_f25_s0	WTBD-group, 25% of train scenes	BladeYOLO-v2, frozen DINOv3 ViT-L/16, seed 0	150	0.438 / 0.219 (epoch 107)	**0.337 / 0.115**; TTA 0.357 / 0.153	Phase 27

Run folder	Data	Model	Epochs	Val best (mAP50 / 50-95)	Test (mAP50 / 50-95)	Notes
`bladeyolo2l_wtbd_group_f25_s1`	WTBD-group, 25% of train scenes	same, seed 1	150	0.484 / 0.235 (epoch 81)	**0.301 / 0.109**; TTA 0.348 / 0.146	Phase 27
`bladeyolo2nodino_wtbd_group_f50`	WTBD-group, 50% of train scenes	BladeYOLO-v2 without DINOv3	150	0.293 / 0.166 (epoch 105)	**0.225 / 0.093**; TTA 0.244 / 0.111	Phase 27
`bladeyolo2nodino_wtbd_group`	WTBD-group (leak-free)	BladeYOLO-v2 without DINOv3	150	0.404 / 0.247 (epoch 108)	**0.361 / 0.170**; TTA 0.385 / 0.210	Phase 27
`yolo12l_wtbd_group`	WTBD-group (leak-free)	YOLO12-L	150	0.419 / 0.269 (epoch 130)	**0.438 / 0.206**; TTA 0.440 / 0.230	Same recipe
`yolo12l_wtbd_official`	WTBD official (**leaky**)	YOLO12-L	120	0.665 / 0.394 (epoch 104)	**0.704 / 0.388**; TTA 0.734 / 0.438	Same recipe

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